Chapter 12
Feedback

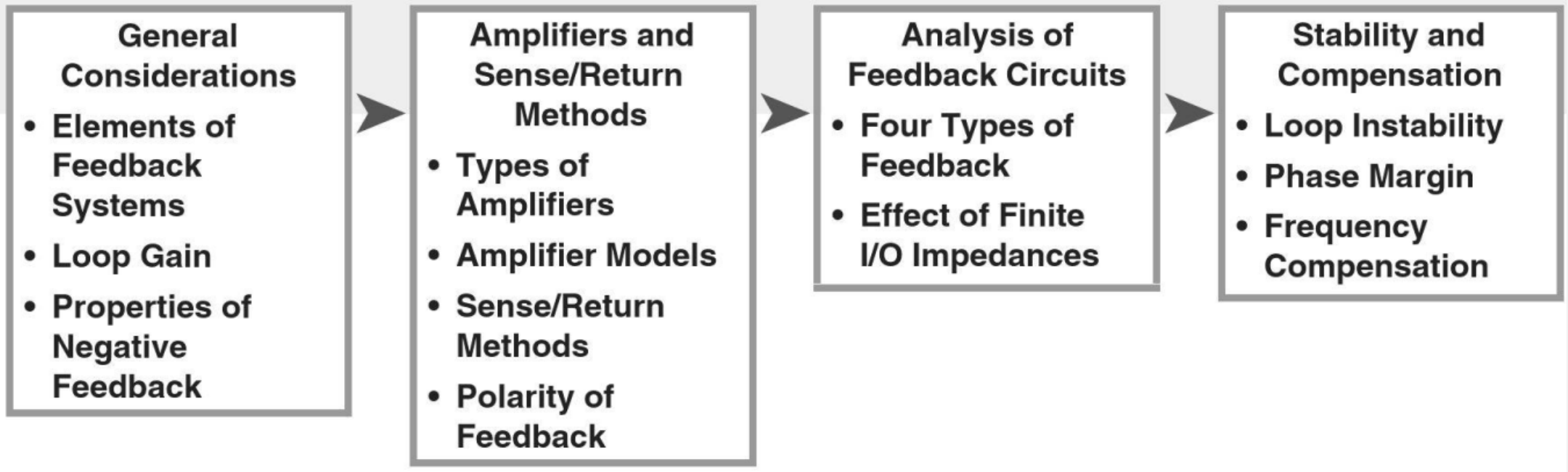
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Ch 12 - Feedback

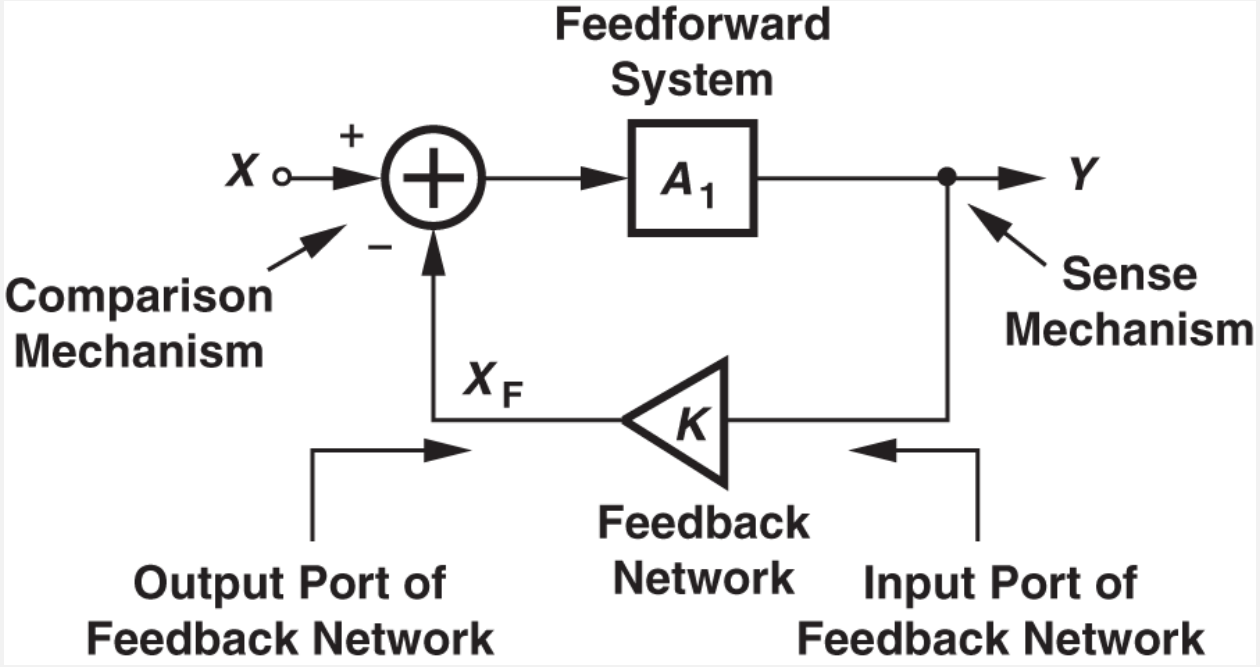
- 12.1 General Considerations
- 12.2 Types of Amplifiers
- 12.3 Sense and Return Techniques
- 12.4 Polarity of Feedback
- 12.5 Feedback Topologies
- 12.6 Effect of Finite I/O Impedances
- 12.7 Stability in Feedback Systems

Chapter Outline

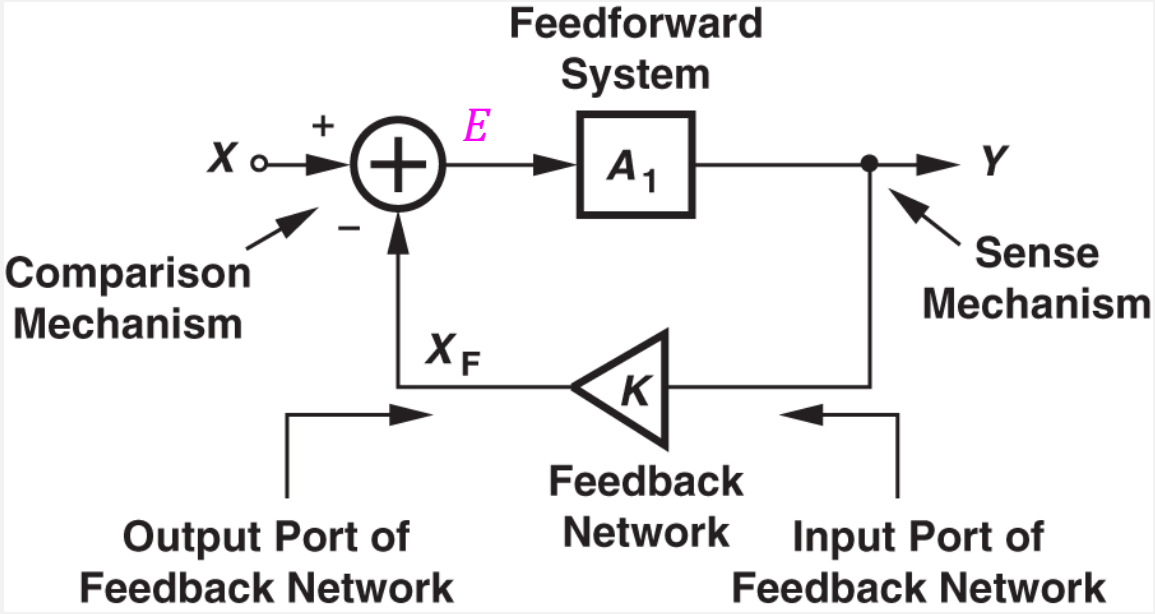


Negative Feedback System

- A negative feedback system consists of four components: 1) feedforward system, 2) sense mechanism, 3) feedback network, and 4) comparison mechanism.



Close-loop Transfer Function



$Y = A_1(X - X_F)$

$X_F = KY$

$Y = A_1(X - KY)$

close-loop transfer function:

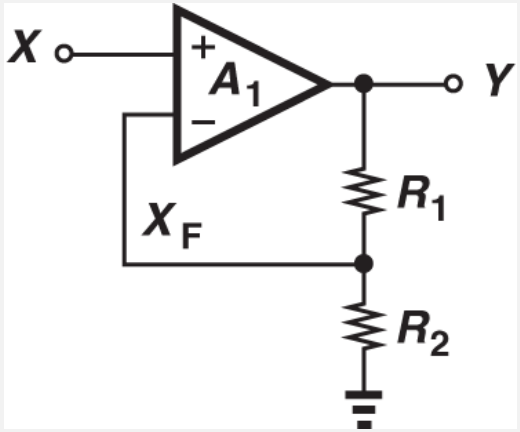
$$H = \frac{Y}{X} = \frac{A_1}{1 + KA_1}$$

error: $E = X - X_F$

feedback factor: $K = \frac{X_F}{Y}$

Example 12-1

- Analyze the noninverting amplifier of Fig. 12.2 from a feedback point of view.



Solution:

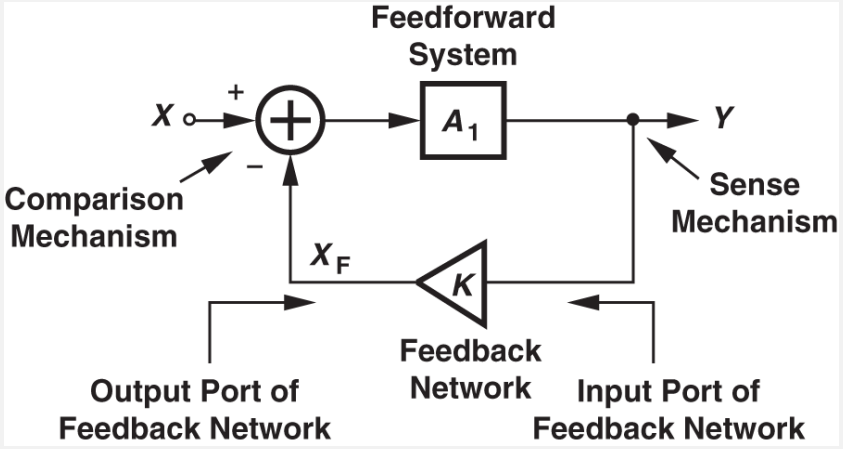
feedback factor:

$$K = \frac{R_2}{R_1 + R_2}$$

$$\frac{Y}{X} = \frac{A_1}{1 + \underbrace{\left(\frac{R_2}{R_1 + R_2}\right)}_{=K} A_1}$$

Comparison Error

- As A_1K increases, the error between the input and fed back signal decreases. Or the fed back signal approaches a good replica of the input.

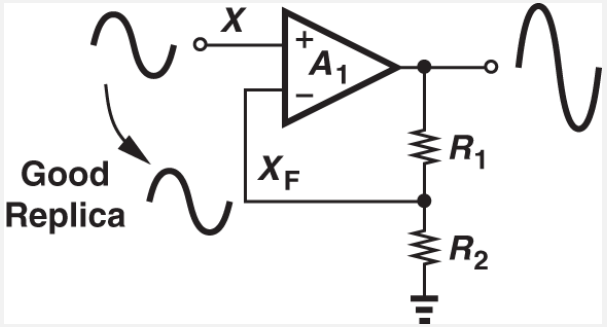


$$E = X - X_F$$
$$X_F = KA_1E$$

$$E = X - KA_1E$$

$$E = \frac{X}{1 + KA_1}$$

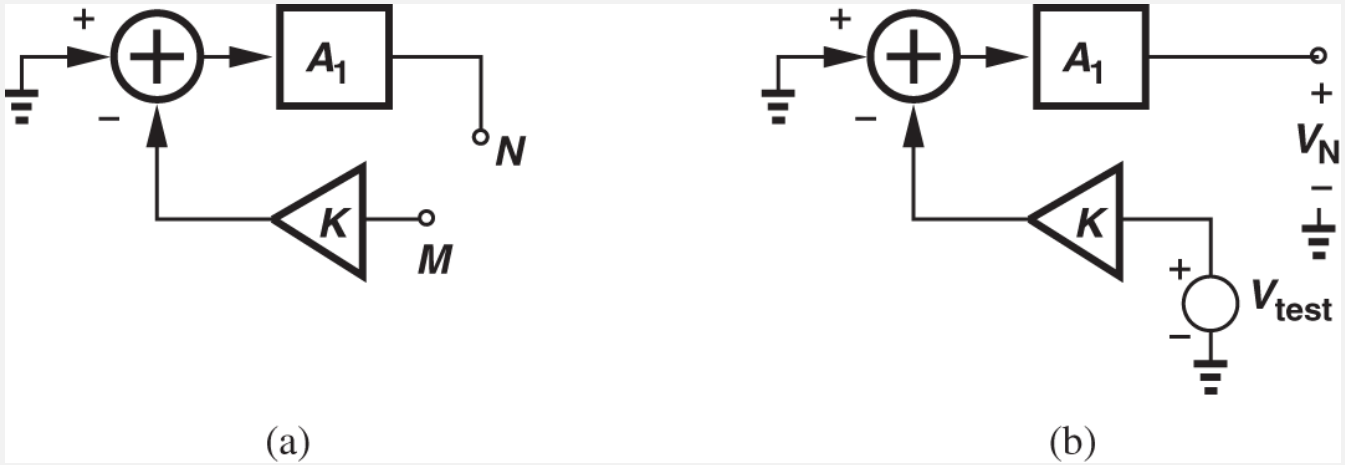
$$KA_1 \uparrow \Rightarrow E \to 0$$
$$\Rightarrow X_F \sim X$$



Feedback signal as a good replica of the input.

Loop Gain

- When the input is grounded, and the loop is broken at an arbitrary location, the loop gain is measured to be KA_1.



$$V_N = -KA_1V_{test}$$

$$KA_1 = -\frac{V_N}{V_{test}}$$

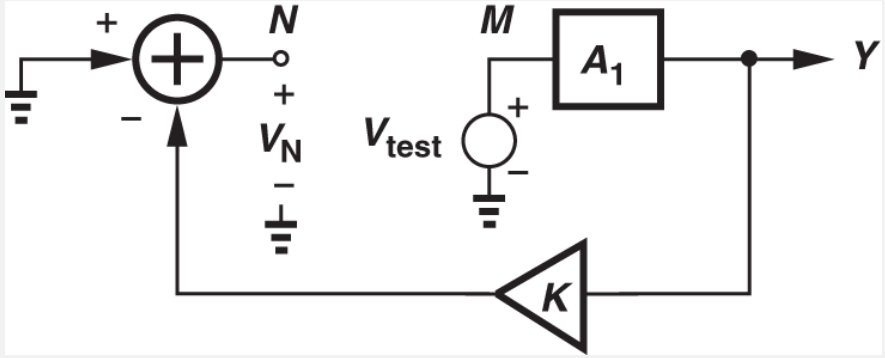
(not to confuse the closed-loop gain)

$$\frac{A_1}{1 + KA_1}$$

Computation of the loop gain by (a) breaking the loop and (b) applying a test signal.

Example 12-3: Alternative Loop Gain Measurement

- Illustrated in Fig. 12.5 is the system with the test signal applied to the input of A_1. The output of the feedback network is equal to KA_1V_{test}, yielding



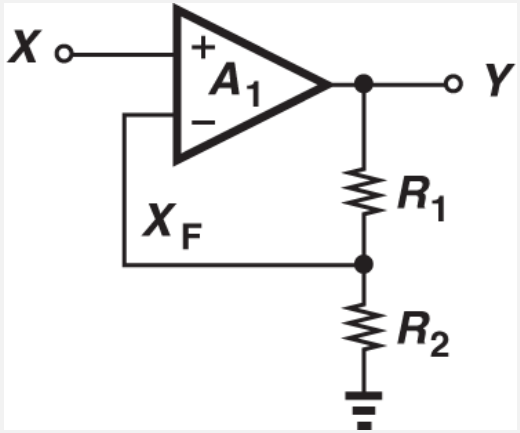
$$V_N = -KA_1V_{test}$$

$$KA_1 = -\frac{V_N}{V_{test}}$$

Gain Desensitization

(PROPERTIES OF NEGATIVE FEEDBACK)

- **Problem**: Suppose A_1 in Fig. 12.1 is an amplifier whose gain is poorly controlled. For example, a CS stage provides a voltage gain of $g_m R_D$ while both g_m and R_D vary with process and temperature; the gain thus may vary by as much as ±20%. Also, suppose we require a voltage gain of 4.00. How can we achieve such precision?
- **Solution**: Equation (12.2) points to a potential solution: if $(KA_1 \gg 1)$, we have



$$K \triangleq \frac{R_2}{R_1 + R_2}$$

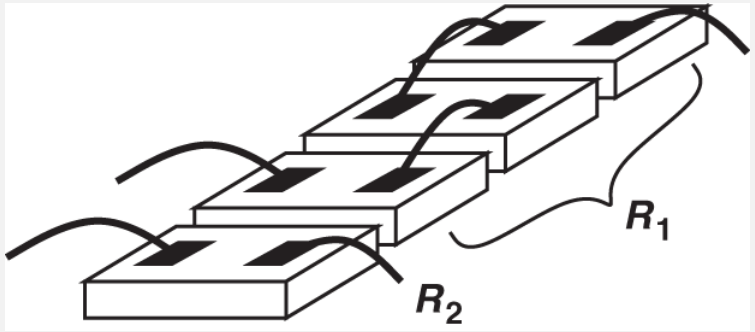
$$\frac{Y}{X} = \frac{A_1}{1 + KA_1}$$

$$\approx \frac{A_1}{KA_1} \quad (KA_1 \gg 1)$$

$$= \frac{1}{K} = 1 + \frac{R_1}{R_2}$$

Ratio of Resistors

- Why is R_1/R_2 more precisely defined than g_mR_D is? If R_1 and R_2 are made of the same material and constructed identically, then the variation of their value with process and temperature does not affect their ratio.
- When two resistors are composed of the same unit resistor, their ratio is very accurate. Since when they vary, they will vary together and maintain a constant ratio.



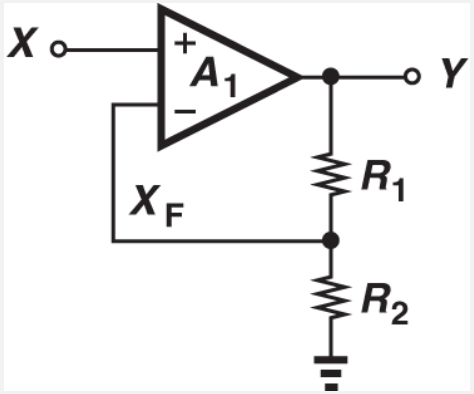
Construction of resistors for good matching.

As an example, for a closed-loop gain of 4.00, we choose

$$R_1 = 3R_2$$

Example 12-4

- The circuit of Fig.12.2 is designed for a nominal gain of 4. (a) Determine the actual gain if A_1 = 1000. (b) Determine the percentage change in the gain if A_1 drops to 500.



Solution:

For a nominal gain of 4,

$$1+\frac{R_1}{R_2}=4 \;\Rightarrow\; \frac{R_1}{R_2}=3$$

$$K=\frac{R_2}{R_1+R_2}=\frac{1}{1+\frac{R_1}{R_2}}=\frac{1}{4}$$

(a) for $A_1 = 1000$,

$$KA_1=\left(\frac{1}{4}\right)1000=250$$

$$\frac{Y}{X}=\frac{A_1}{1+KA_1}=\frac{1000}{1+250}=3.984$$

(b) for $A_1 = 500$,

$$KA_1=\left(\frac{1}{4}\right)500=125$$

$$\frac{Y}{X}=\frac{A_1}{1+KA_1}=\frac{1000}{1+125}=3.968$$

A_1 drops by factor of 2, the closed-loop gain changes by only

$$\frac{(3.984-3.968)}{3.984}=0.4\%$$

Merits of Negative Feedback

- The above example reveals that the closed-loop gain of a feedback circuit becomes relatively independent of the open-loop gain so long as the loop gain, KA_1 remains sufficiently higher than unity. This property of negative feedback is called "gain desensitization."
- We now see why we are willing to accept a reduction in the gain by a factor of $1+KA_1$. We begin with an amplifier having a high, but poorly-controlled gain and apply negative feedback around it so as to obtain a better-defined, but inevitably lower gain.
- The gain desensitization property of negative feedback means that any factor that influences the open-loop gain has less effect on the closed-loop gain. Thus far, we have blamed only process and temperature variations, but many other phenomena change the gain as well.
  - **Bandwidth enhancement**
  - **Modification of I/O Impedances**
  - **Linearization**

Bandwidth Enhancement

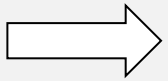
- Let us consider a one-pole open-loop amplifier with a transfer function. Although negative feedback lowers the gain by ($1+KA_0$), it also extends the bandwidth by the same amount.

Open Loop

$$A_1(s) = \frac{A_0}{1 + \frac{s}{\omega_0}}$$

Open-Loop Gain = A_0
Open-Loop Bandwidth = ω_0

Negative Feedback



Closed Loop

$$\frac{Y}{X}(s) = \frac{A_1}{1 + KA_1} = \frac{\frac{A_0}{1+\frac{s}{\omega_0}}}{1 + K\frac{A_0}{1+\frac{s}{\omega_0}}}$$

$$= \frac{\frac{A_0}{1 + KA_0}}{1 + \frac{s}{(1 + KA_0)\omega_0}}$$

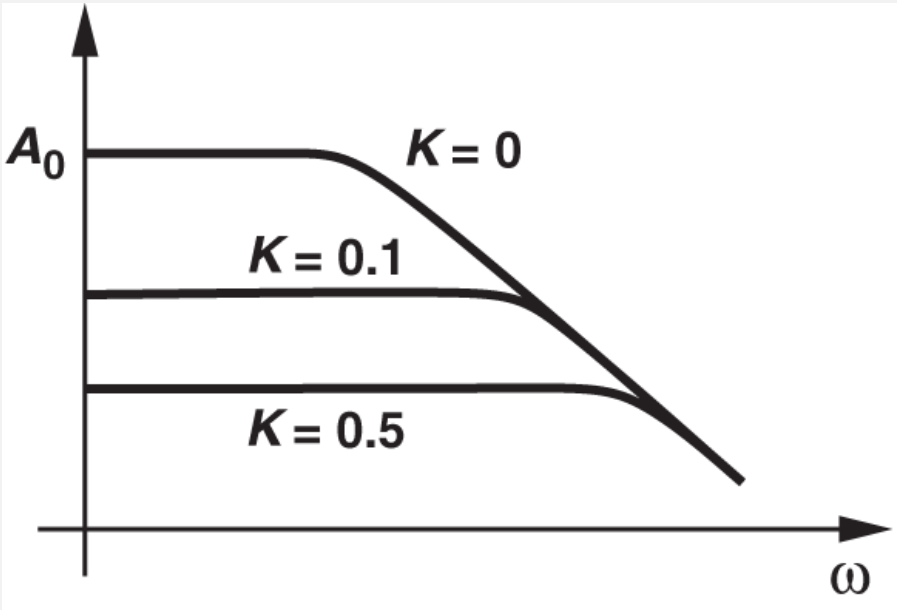
Closed-Loop Gain = $\frac{A_0}{1+KA_0}$
Closed-Loop Bandwidth = $(1 + KA_0)\omega_0$

Example 12-5

- Plot the closed-loop frequency response given by Eq. (12.22) for K = 0, 0.1, and 0.5. Assume A_0 = 200.

Solution:

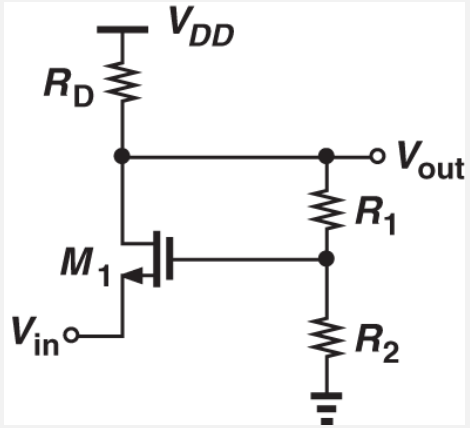
$$\frac{Y}{X}(s) = \frac{\frac{A_0}{1+KA_0}}{1+\frac{s}{(1+KA_0)\omega_0}} \qquad (12.22)$$



$K = 0$: $1 + KA_0 = 1$
$K = 0.1$: $1 + KA_0 = 1 + 0.1\times200 = 21$
$K = 0.5$: $1 + KA_0 = 1 + 0.5\times200 = 101$

Example 12-7: Transistor-level Realization of the Feedback Circuit

- Figure 12.11 depicts a transistor-level realization of the feedback circuit shown in Fig. 12.2. Assume $\lambda = 0$ and $R_1 + R_2 \gg R_D$ for simplicity, (a) Identify the four components of the feedback system, (b) Determine the open-loop and closed-loop voltage gain, (c) Determine the open-loop and closed-loop I/O impedances.



Review of CG Stage

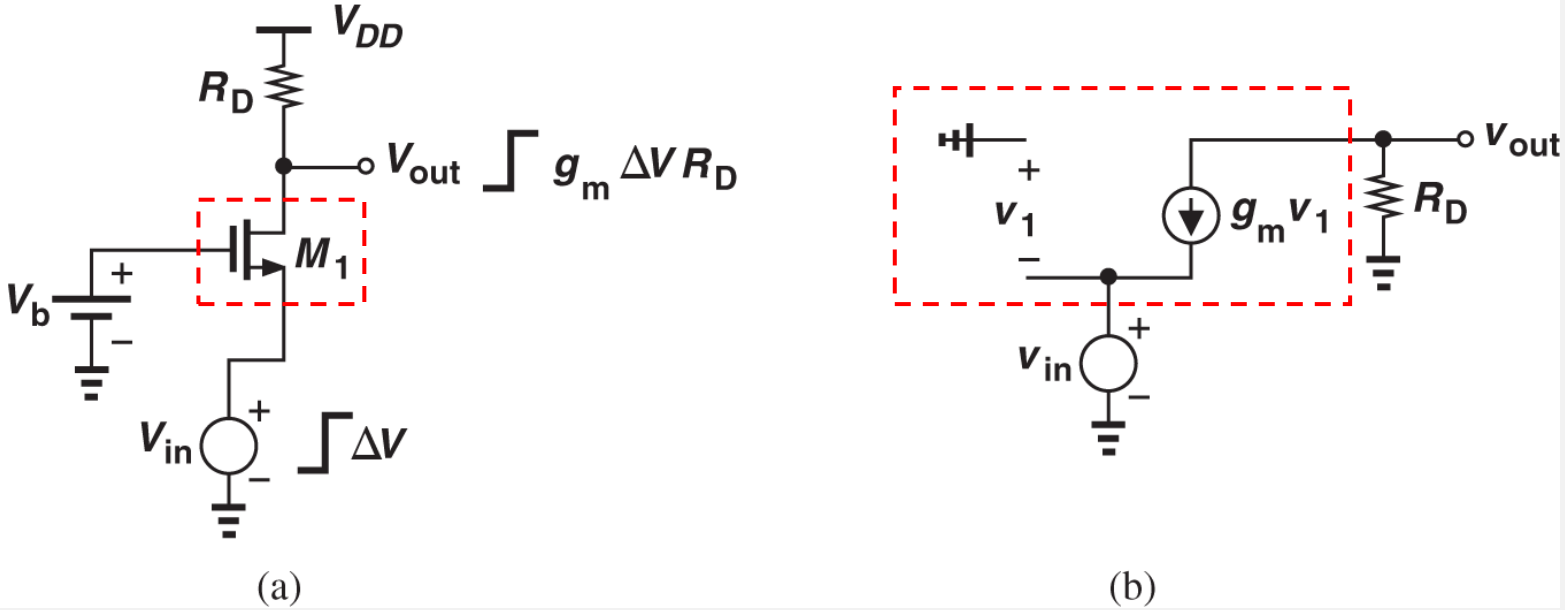
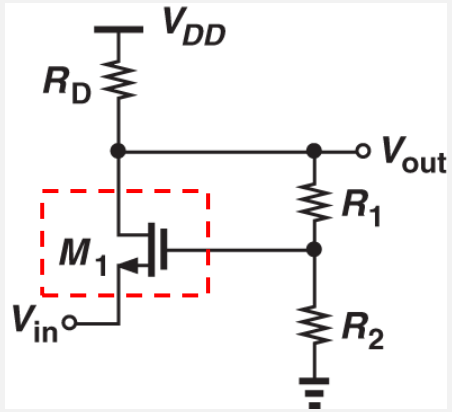


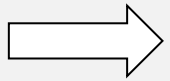
Figure 17.50 (a) Response of CG stage to small input change, (b) small-signal model.

Example 12-7(a): Feedback Viewpoint

Solution:

(a) Identify the four components of the feedback system

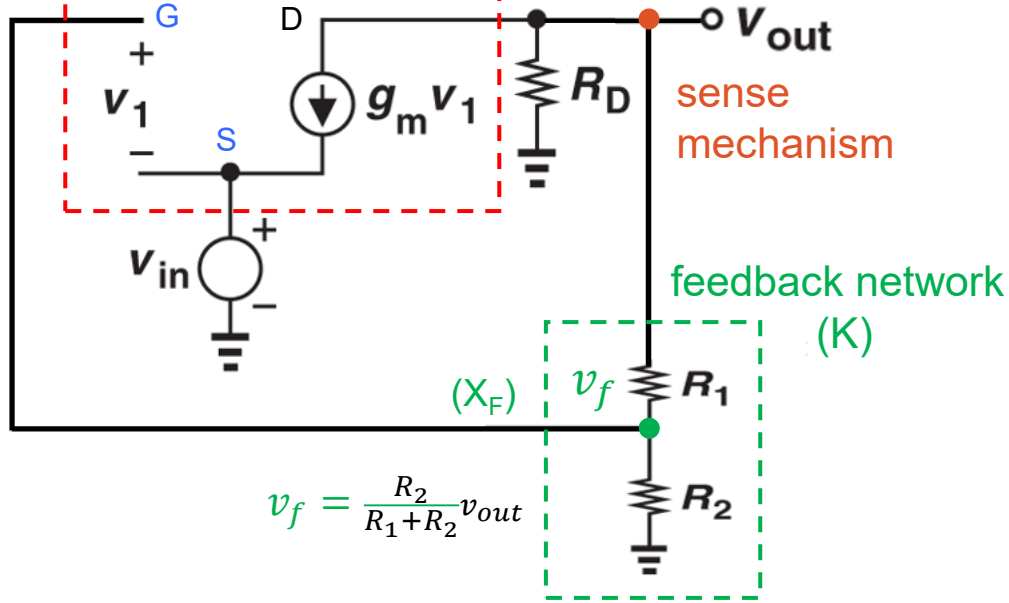




feedforward system (A_0)

comparison mechanism

$$v_{out} = -(g_m v_1)R_L$$
$$= g_m R_L v_{SG}$$
$$= g_m R_L (v_S - v_G)$$
$$= \underbrace{g_m R_L}_{A_0}(v_{in} - v_f)$$



- transistor M_1 itself operates as the subtractor.
- $R_L = R_D||(R_1 + R_2)$

Example 12-7(b): Closed-loop Gain

(b) Determine the open-loop and closed-loop voltage gain

open-loop gain:

$$R_L = R_D||(R_1 + R_2)$$
$$\approx R_D \quad (\because R_1 + R_2 \gg R_D)$$

$$v_{out} \approx -(g_m v_1) R_D$$
$$= g_m R_D v_{in}$$
$$A_0 \approx g_m R_D$$

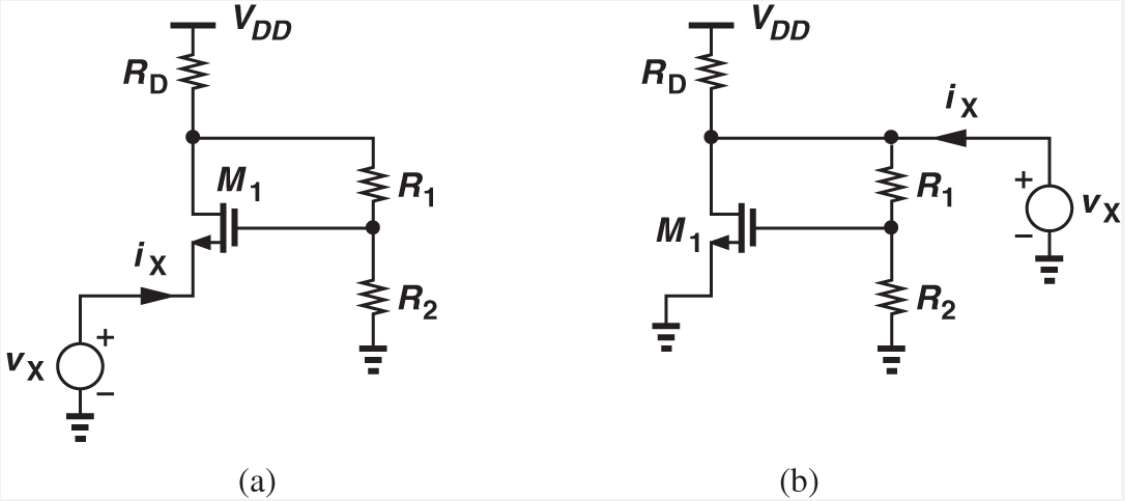
close-loop gain:

$$K = \frac{v_f}{v_{out}} = \frac{R_2}{R_1 + R_2}$$

$$\frac{v_{out}}{v_{in}} = \frac{A_0}{1 + KA_0} = \frac{g_m R_D}{1 + \frac{R_2}{R_1 + R_2} g_m R_D}$$

Example 12-7(c): I/O Impedances Improvement

(c) Determine the open-loop and closed-loop I/O impedances.

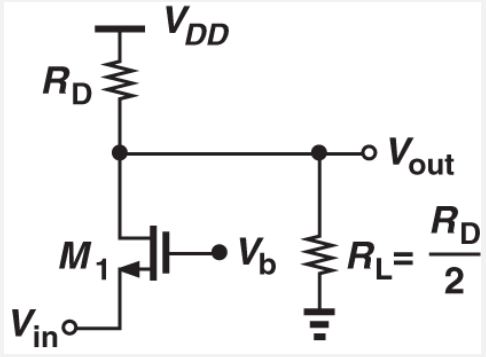


$$R_{in} = \frac{1}{g_m}(1 + KA_0)$$
$$= \frac{1}{g_m}\left(1 + \frac{R_2}{R_1 + R_2} g_m R_D\right)$$

$$R_{out} = \frac{R_D}{1 + KA_0}$$
$$= \frac{R_D}{1 + \frac{R_2}{R_1 + R_2} g_m R_D}$$

Example 12-8: Load Desensitization

- The common-gate stage of Fig.12.11 must drive a load resistance $R_L = R_D/2$.How much does the gain change (a) without feedback, (b) with feedback?



Solution:

(a) without feedback:

$$\frac{v_{out}}{v_{in}} = g_m(R_D||R_L) = g_m\left(R_D||\frac{R_D}{2}\right) = \frac{g_m R_D}{3}$$

That is, the gain drops by factor of three.

(b) with feedback:

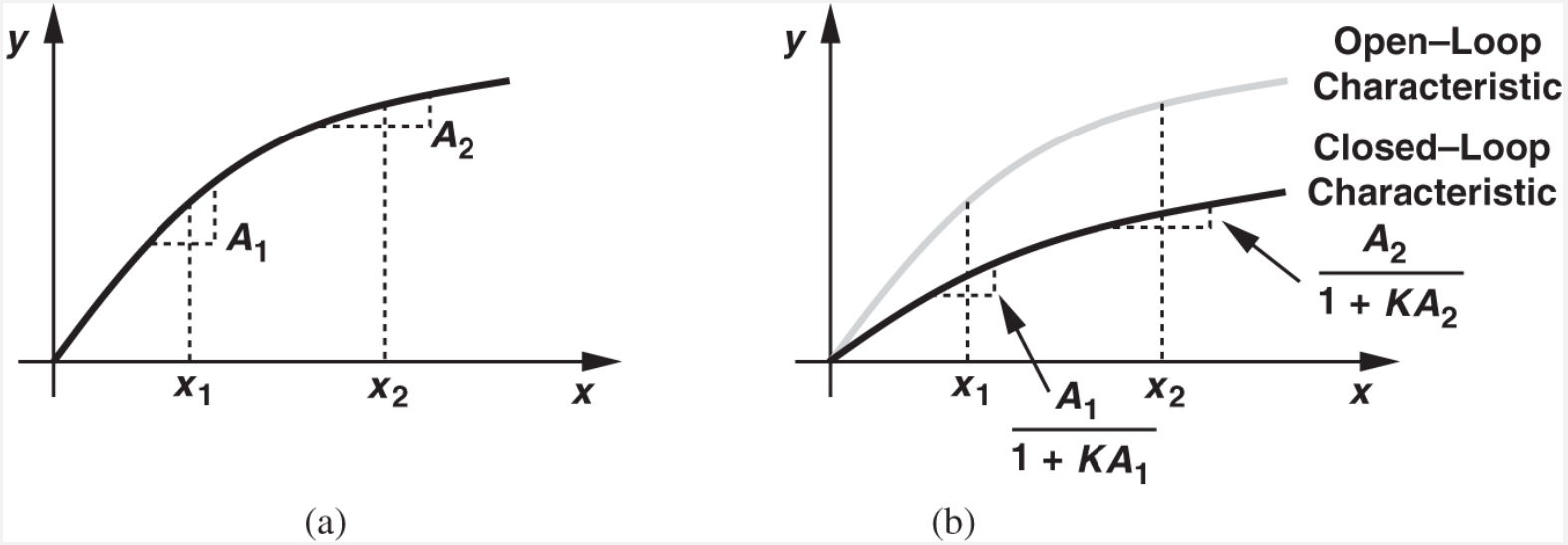
$$K = \frac{v_f}{v_{out}} = \frac{R_2}{R_1+R_2}$$

$$\frac{v_{out}}{v_{in}} = \frac{A_0}{1+KA_0} = \frac{g_m R_D/3}{1+\frac{R_2}{R_1+R_2}(g_m R_D/3)} = \frac{g_m R_D}{3+\frac{R_2}{R_1+R_2}g_m R_D}$$

For example, if $\frac{R_2}{R_1+R_2}g_m R_D = 10$, then this result differs from the "unloaded" by about 18%.

Linearity Improvement

- If placed in a negative-feedback loop, the system provides a more uniform gain for different signal levels.



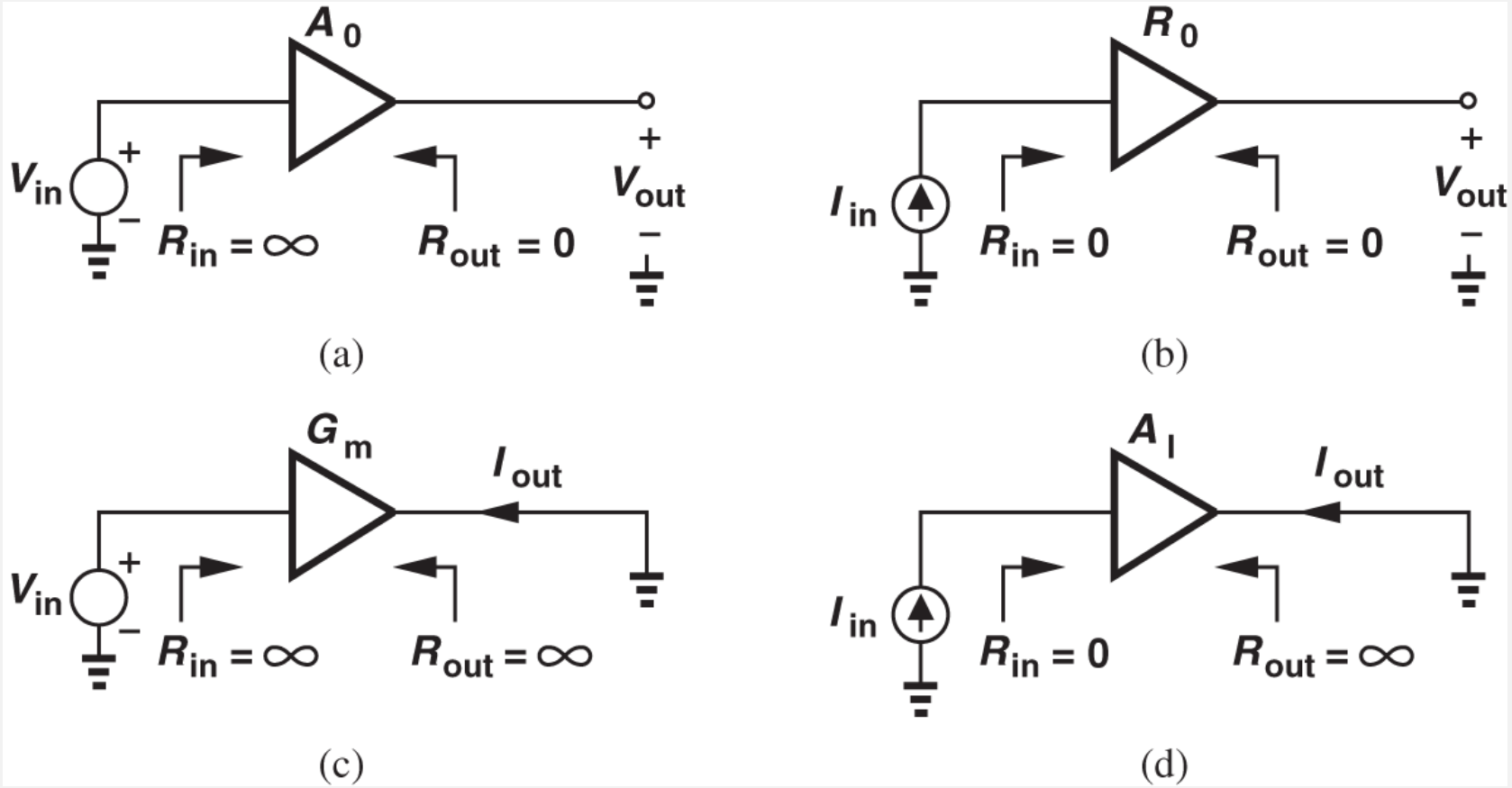
gain at x_i:

$$\frac{A_i}{1 + KA_i} = \frac{1}{K}\left(1 - \frac{1}{KA_i}\right)$$

$$\approx \frac{1}{K} \quad (\text{if } KA_i \gg 1)$$

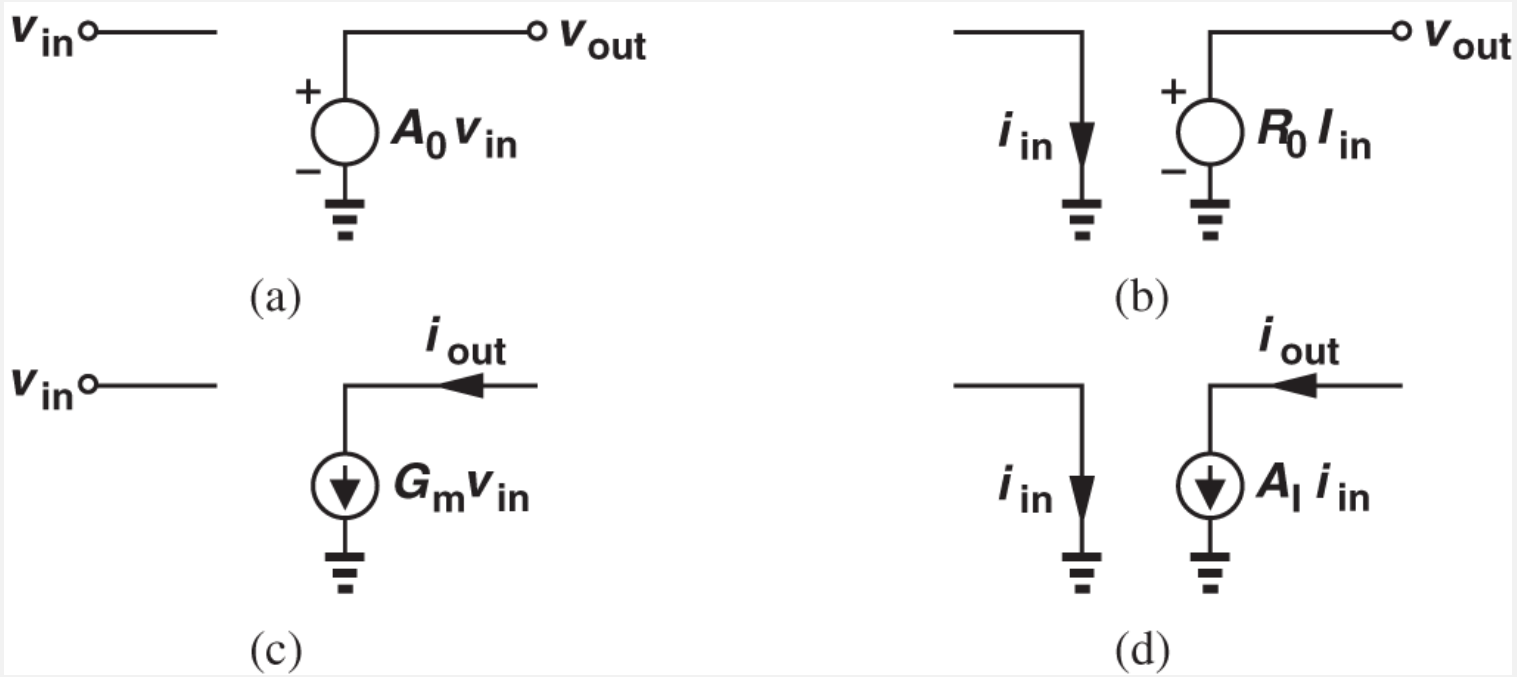
(a) Nonlinear open-loop characteristic of an amplifier,
(b) improvement in linearity due to feedback.

Four Types of Amplifiers



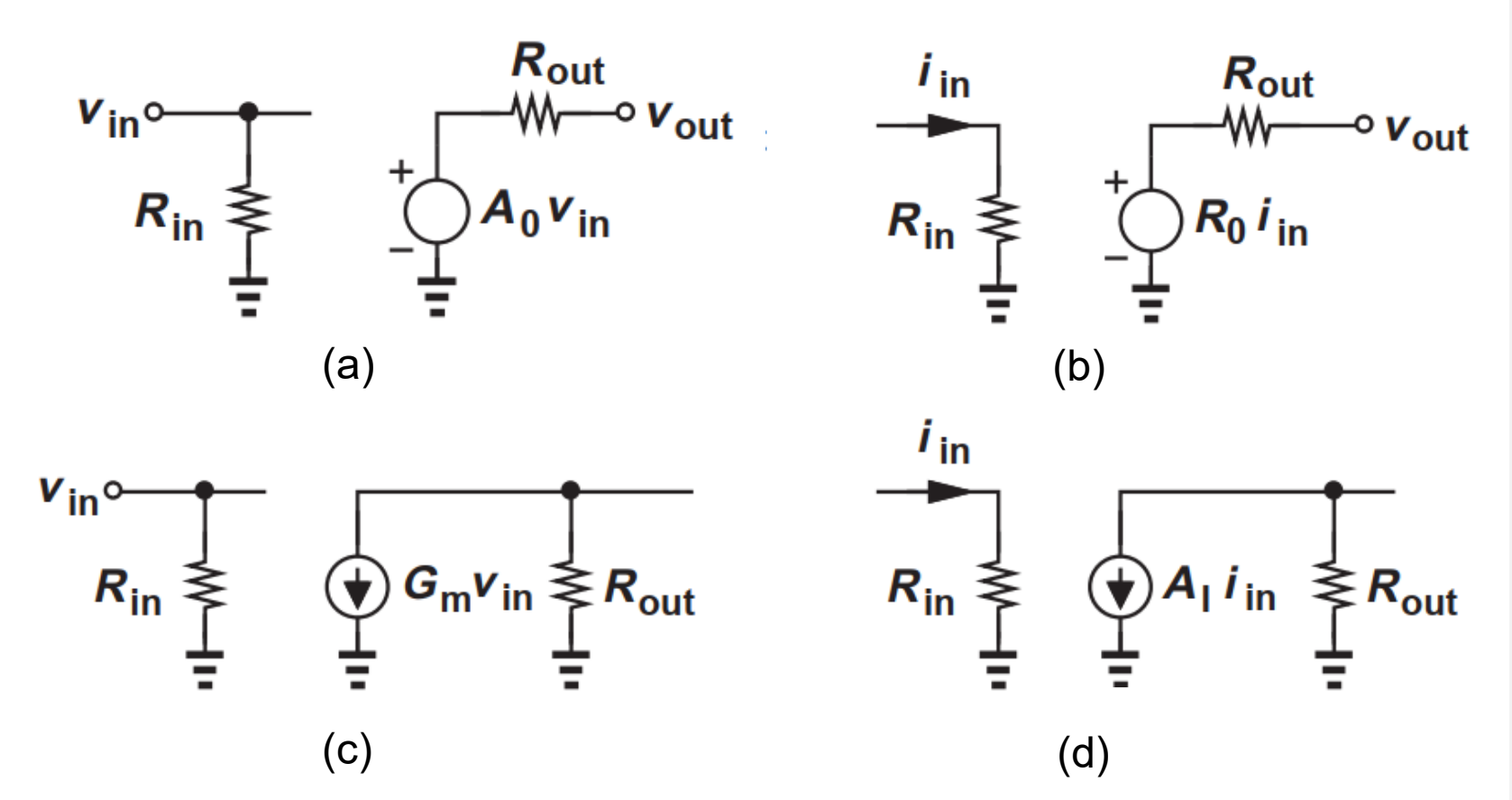
(a) voltage, (b) transimpedance, (c) transconductance, and (d) current amplifiers

Ideal Model of Amplifiers



(a) voltage, (b) transimpedance, (c) transconductance, and (d) current amplifiers

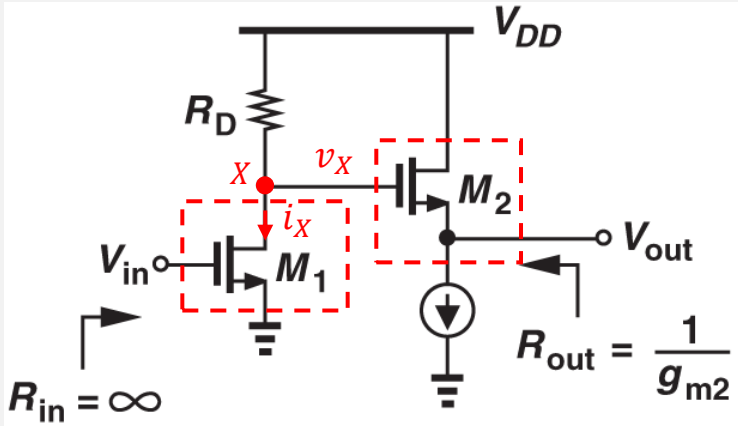
Realistic Model of Amplifiers



(a) voltage, (b) transimpedance, (c) transconductance, and (d) current amplifiers

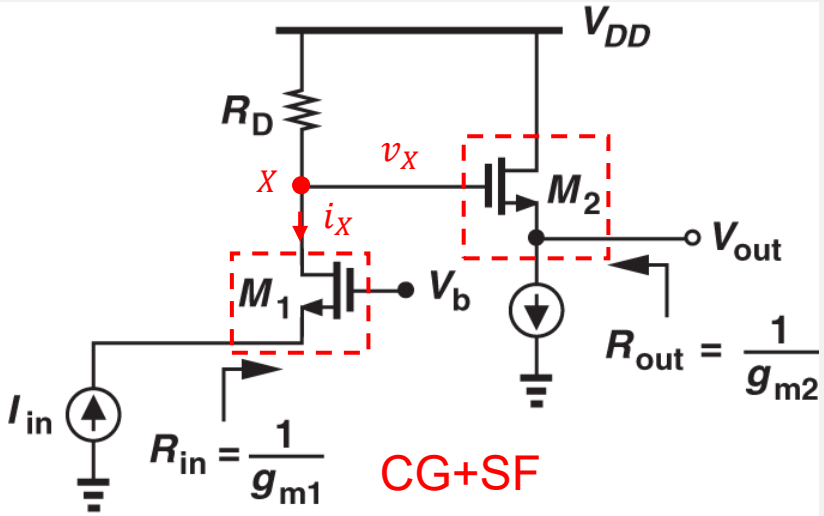
Examples of the Four Amplifier Types

$$A_V = \frac{v_{out}}{v_{in}} = -g_m R_D$$



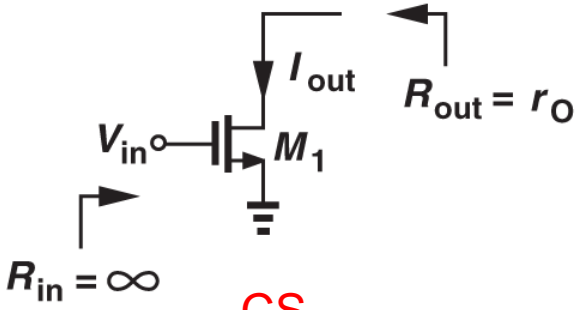
(a)

$$R_T = \frac{v_{out}}{i_{in}} = R_D$$



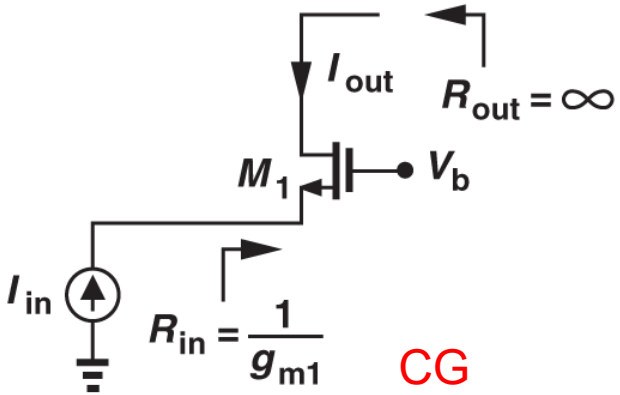
(b)

$$G_m = \frac{i_{out}}{v_{in}} = g_m$$



(c)

$$A_I = \frac{i_{out}}{i_{in}} = 1$$

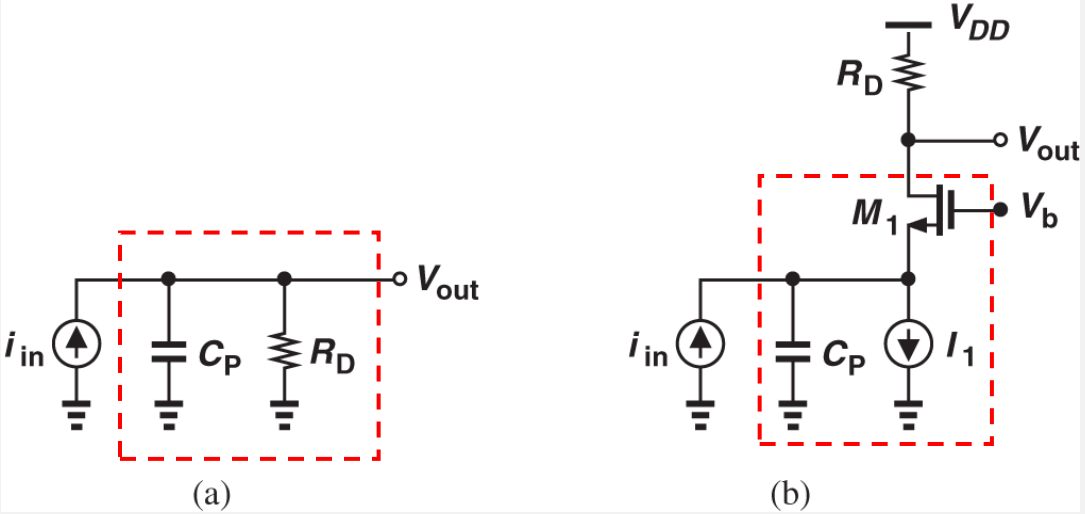


(d)

Examples of (a) voltage, (b) transimpedance, (c) transconductance, and (d) current amplifiers.

Example 12-9

- With a current gain of unity, the CG topology of Fig. 12.19(d) appears hardly better than a piece of wire. What is the advantage of this circuit?



Solution:

The important property of this circuit lies in its input impedance. Suppose the current source serving as the input suffers from a large parasitic capacitance, C_p.
The -3 dB bandwidth are

$$\omega_{p,a} = \frac{1}{R_D C_p},$$

and

$$\omega_{p,b} = \frac{1}{\left(\frac{1}{g_m}\right) C_p},$$

respectively.

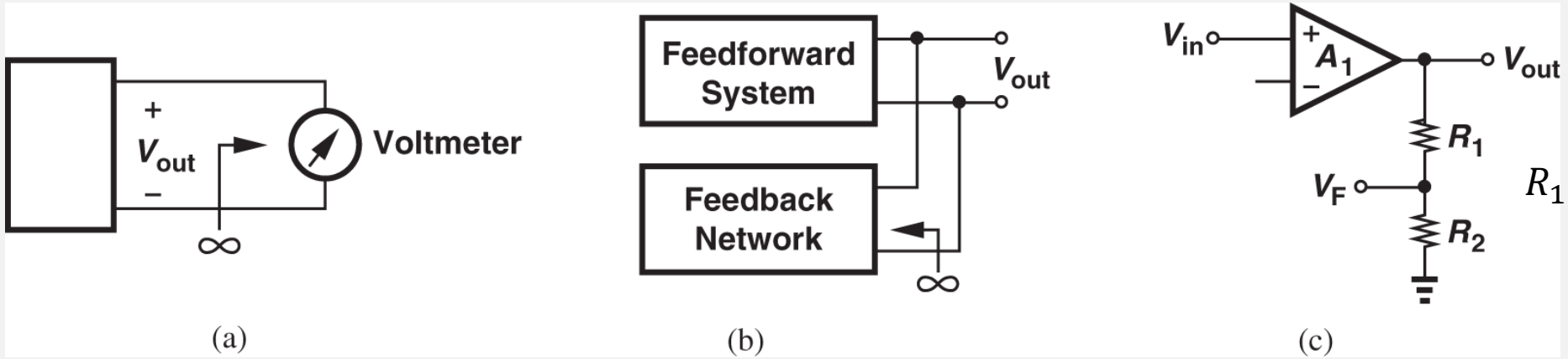
$\because \frac{1}{g_m} < R_D,$

$$\omega_{p,b} > \omega_{p,a}$$

The use of a CG stage moves the input pole to g_m/C_p, a much higher frequency.

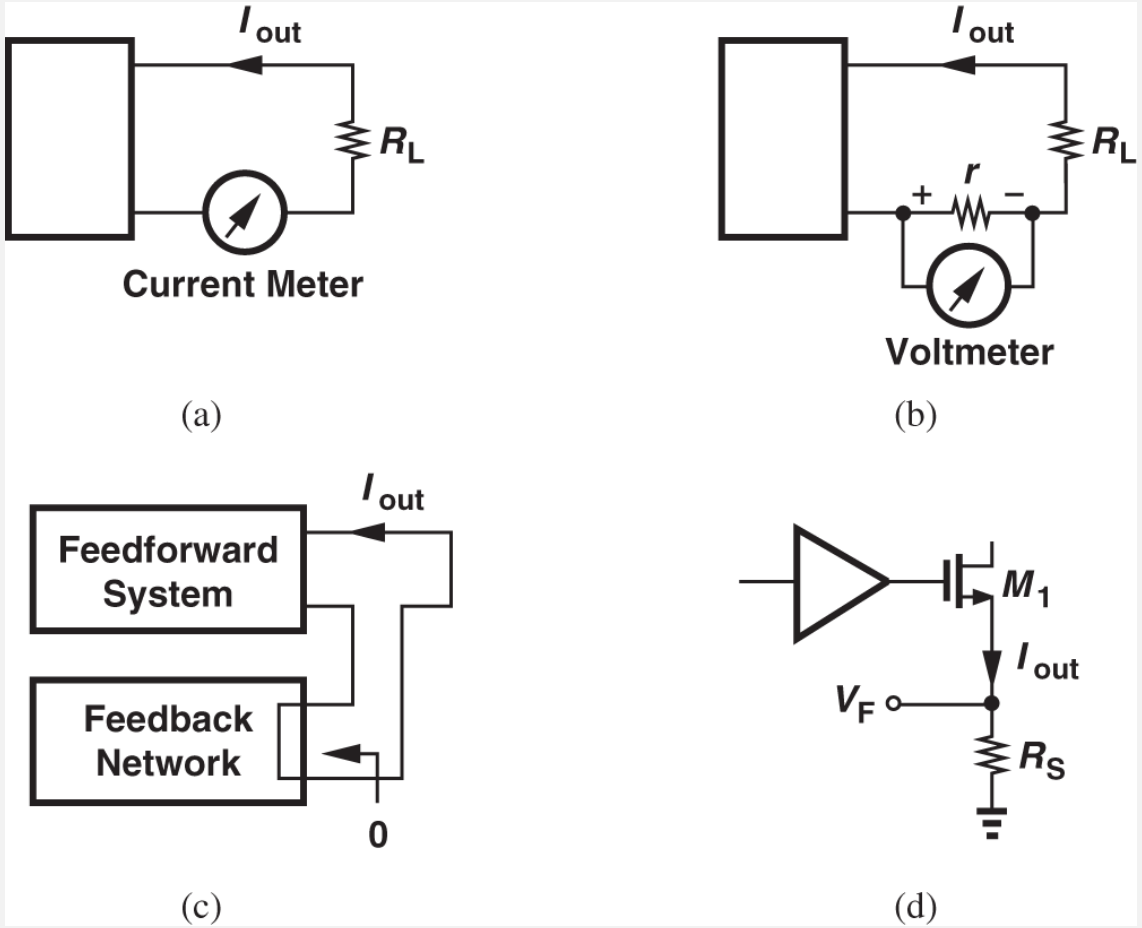
SENSE AND RETURN TECHNIQUES - Voltage

- Recall from Section 12.1 that a feedback system includes means of sensing the output and “returning" the feedback signal to the input. In this section, we study such means so as to recognize them easily in a complex feedback circuit.



(a) Sensing a voltage by a voltmeter, (b) sensing the output voltage by the feedback network, (c) example of implementation.

SENSE AND RETURN TECHNIQUES - Current



(a) Sensing a current by a current meter, (b) actual realization of current meter, (c) sensing the output current by the feedback network, (d) example of implementation.

How do we measure the current flowing through a wire? We *break* the wire and place a current meter in series with the wire, and

$$r \ll R_L,$$

it does not disturb the circuit. and a voltmeter that measures the voltage drop across the resistor.

(d) is an implementation of this concept

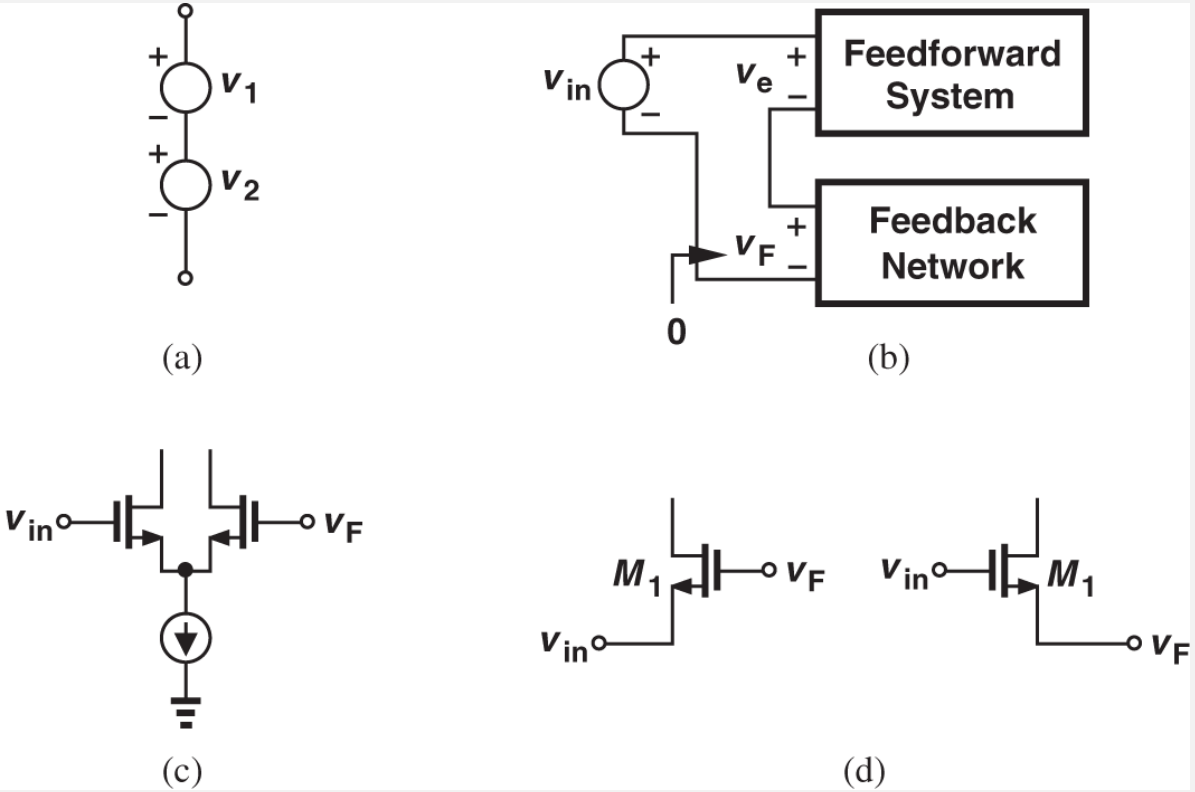
$$R_S \ll \frac{1}{g_{m1}},$$

where $(r = R_S) \ll (R_L = \frac{1}{g_{m1}})$.

$R_S \approx 0$

Comparison Mechanism - Voltage Subtractor

- **Comparison Mechanism**: To return a voltage or current to the input, we must employ a mechanism for adding or subtracting such quantities.
- To add two voltage sources, we place them in series.



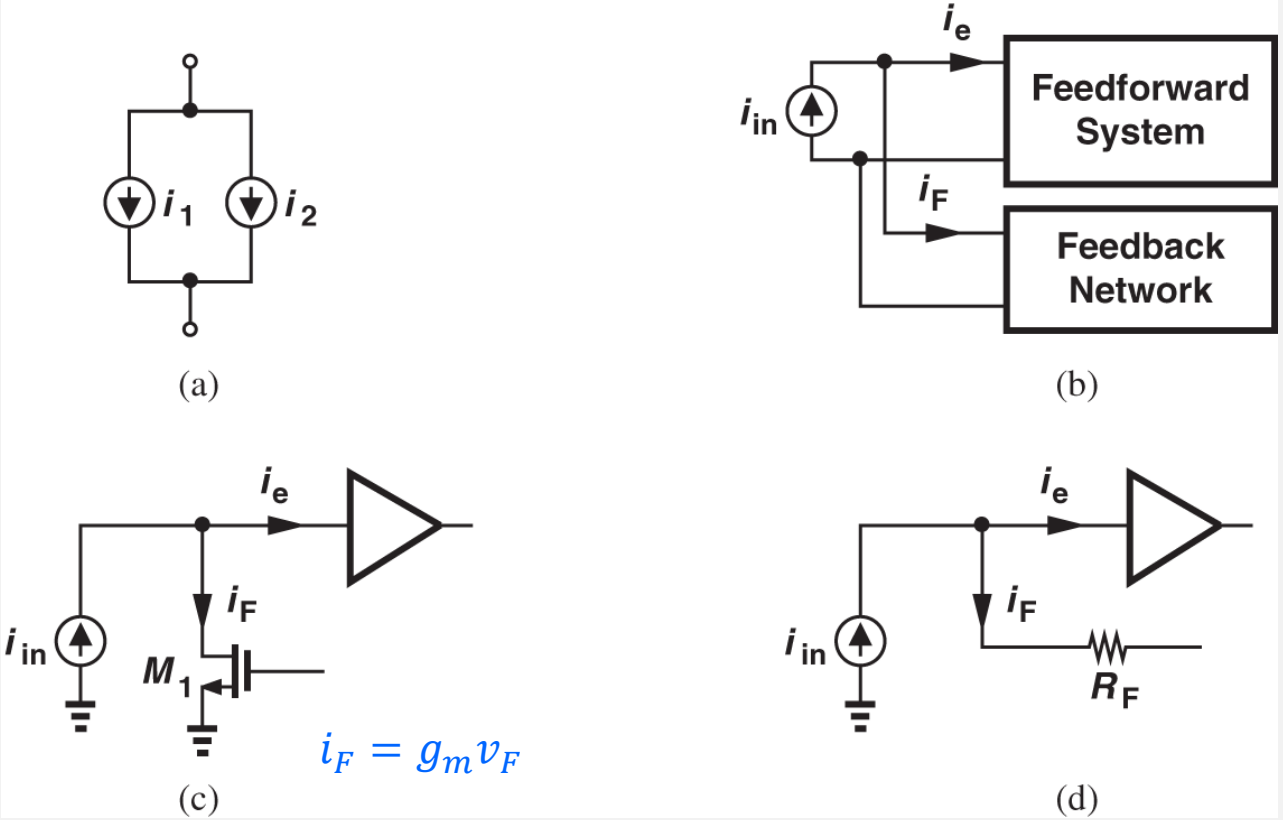
(a) addition of two voltages,
(b) addition of feedback and input voltages,

$$v_{in} = v_e + v_F$$
$$v_e = v_{in} - v_F$$

(c) differential pair as a voltage subtractor,
(d) single transistor as a voltage subtractor.

Comparison Mechanism - Current Subtractor

- To add two current sources, we place them in parallel.



(transconductance amplifier)

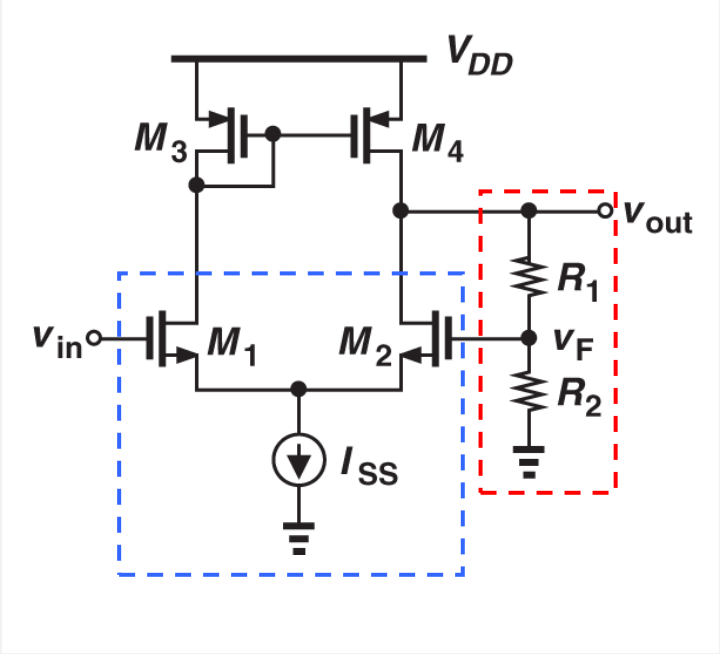
(a) Addition of two currents,
(b) addition of feedback current and input current,

$$i_{in} = i_e + i_F$$
$$i_e = i_{in} - i_F$$

(c) circuit realization: a transistor can return a current to the input,
(d) another realization: a resistor R_F if it is large enough to approximate a current source.

Example 12-10: Voltage-Voltage Feedback

- Determine the types of sensed and returned signals in the circuit of Fig. 12.25.



Solution:

- The resistive divider senses the output voltage and serves as the feedback network

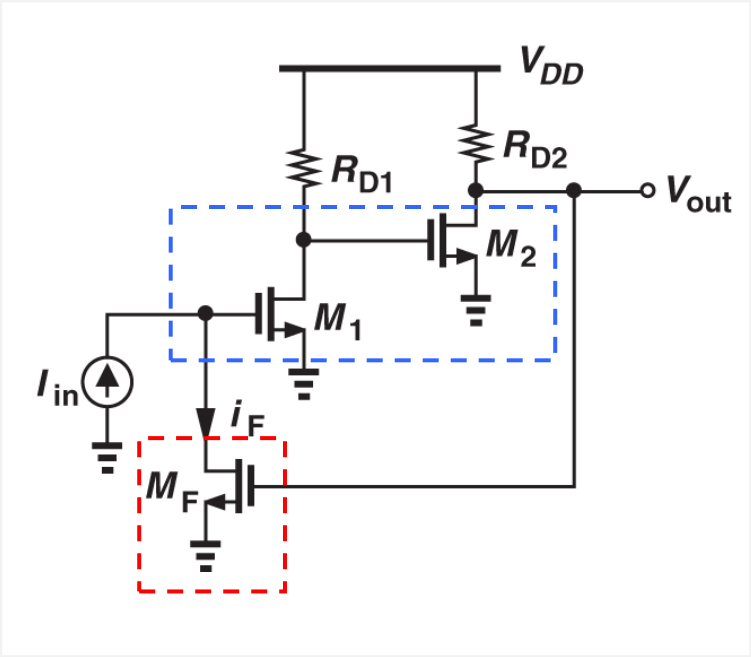
$$v_F = \frac{R_2}{R_1+R_2} v_{out}.$$

- M_1 and M_2 operate as as both part of the op amp (the forward system) and a voltage subtractor.
- The amplifier therefore combines the topologies in Figs. 12.21(c) and 12.23(c).

voltage sense and return: 12.21(c)
voltage subtractor: 12.23(c)

Example 12-11: Voltage-Current Feedback

- Compute the feedback factor, K, for the circuit depicted in Fig. 12.26. Assume $\lambda = 0$.



Solution:

Transistor M_F both senses the output voltage and returns a current to the input.

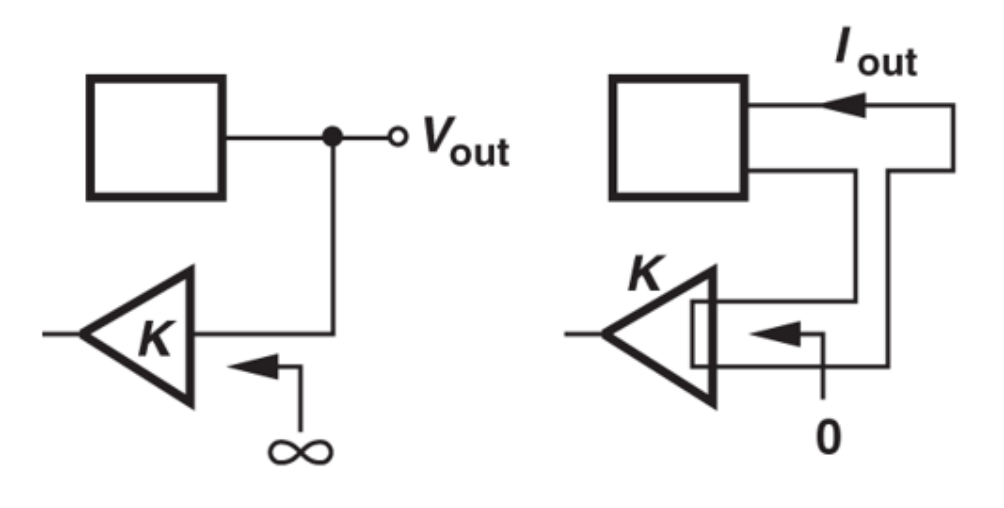
$$i_F = K v_{out}$$

$$K = \frac{i_F}{v_{out}} = g_{mF}$$

where g_{mF} denotes the transconductance of M_F.

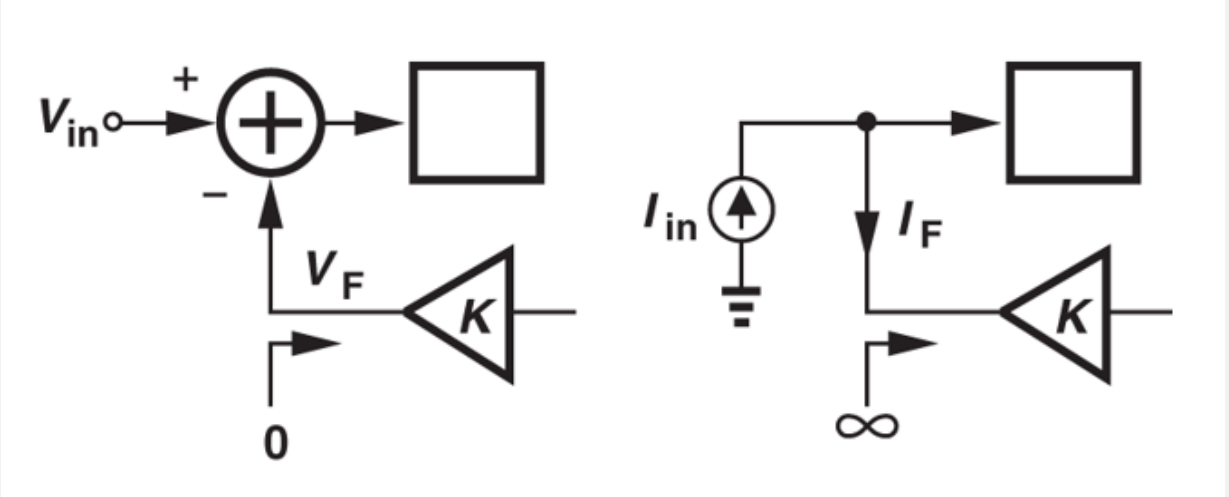
Input Impedance of an Ideal Feedback Network

- To sense a voltage, the input impedance of an ideal feedback network must be infinite.
- To sense a current, the input impedance of an ideal feedback network must be zero.



Output Impedance of an Ideal Feedback Network

- To return a voltage, the output impedance of an ideal feedback network must be zero.
- To return a current, the output impedance of an ideal feedback network must be infinite.



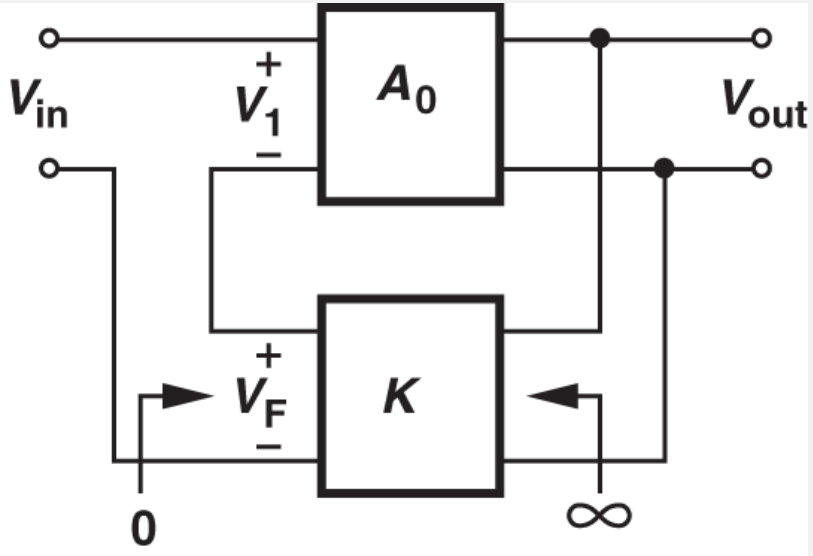
FEEDBACK TOPOLOGIES - Voltage-Voltage Feedback

- **Voltage-Voltage Feedback** (Razavi): (Output-Input)

  In our terminology, the first term in "voltage-voltage feedback" refers to the quantity sensed at the output, and the second, to the quantity returned to the input. (This terminology is not standard.)

- **Series-Shunt Feedback** (Sedra/Smith): (Input-Output)

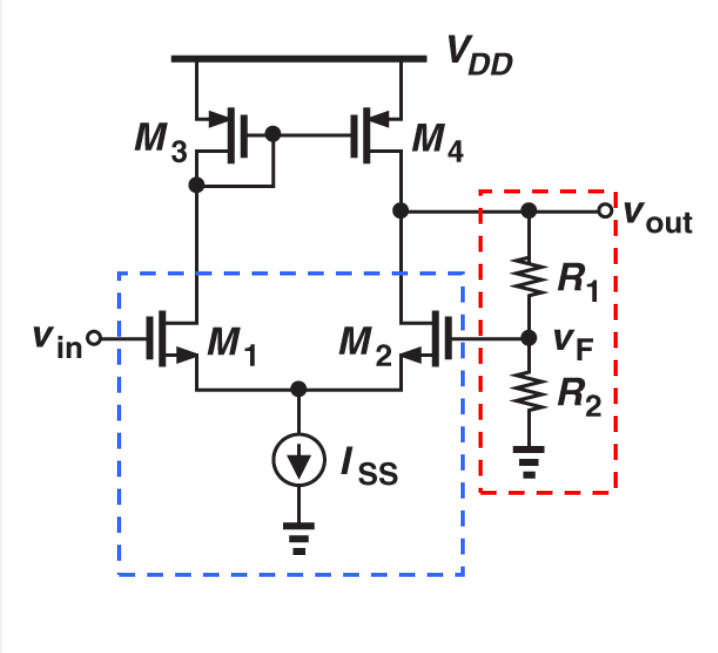
  The first term refers to the return mechanism at the input and the second term to the sense mechanism at the output.



$$V_1 = V_{in} - V_F$$
$$V_{out} = A_0 V_1$$
$$V_F = K V_{out}$$
$$V_{out} = A_0(V_{in} - K V_{out})$$
$$\frac{V_{out}}{V_{in}} = \frac{A_0}{1 + K A_0}$$

Example 12-15: Closed-loop Gain of a V-V Feedback

- Determine the closed-loop gain of the circuit shown in Fig. 12.32, assuming R_1+R_2 is very large.



Solution:

The resistive divider senses the output voltage and serves as the feedback network.

$$v_F = \frac{R_2}{R_1+R_2} v_{out}$$

$$K = \frac{v_F}{v_{out}} = \frac{R_2}{R_1 + R_2}$$

M_1 and M_2 operate as as both part of the op amp (the forward system) and a voltage subtractor.

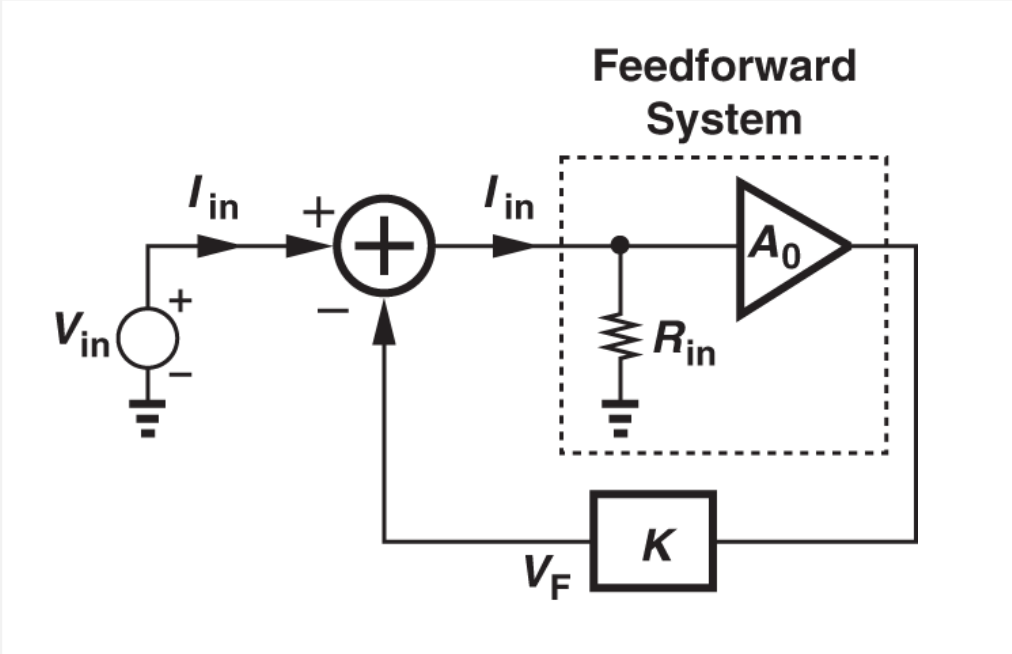
$$A_0 = g_{mN}(r_{ON}||r_{OP})$$

The closed-loop gain is

$$\frac{V_{out}}{V_{in}} = \frac{g_{mN}(r_{ON}||r_{OP})}{1 + \frac{R_2}{R_1 + R_2} g_{mN}(r_{ON}||r_{OP})}$$

Input Impedance of a V-V Feedback

- A better voltage sensor: the negative feedback around a voltage amplifier raises the input impedance by the universal factor of one plus the loop gain. This impedance modification brings the circuit closer to an ideal voltage amplifier.

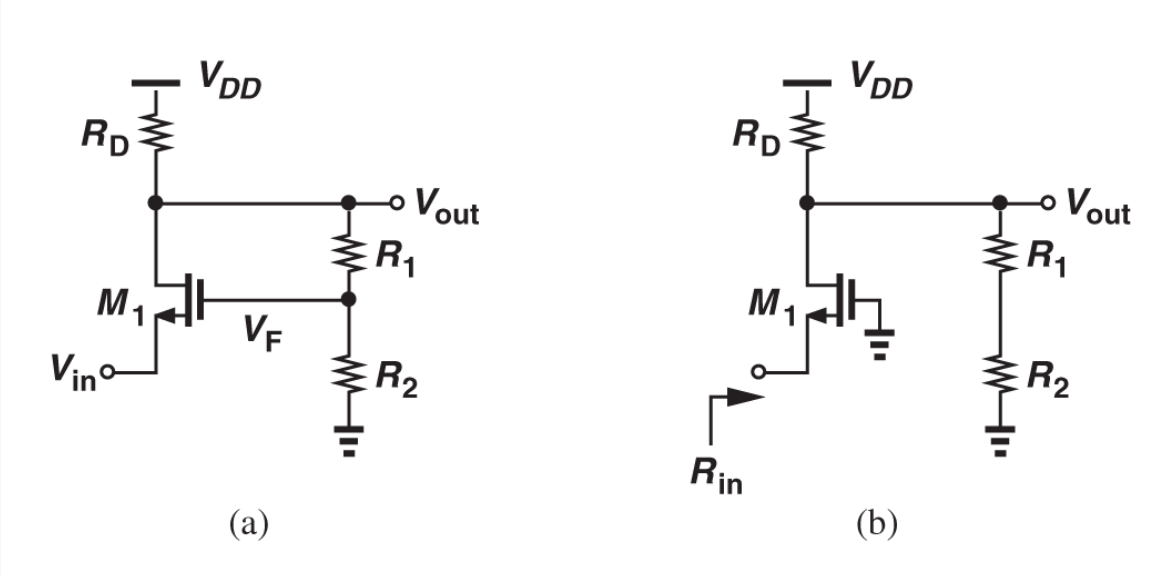


$$I_{in}R_{in} = V_{in} - V_F$$
$$= V_{in} - KA_0I_{in}R_{in}$$

$$\frac{V_{in}}{I_{in}} = R_{in}(1 + KA_0)$$

Example 12-16: Input Impedance of a V-V Feedback

- Determine the input impedance of the stage shown in Fig. 12.34(a) if R_1+R_2 is very large.



Solution:

$$R_{in} = \frac{1}{g_m}$$

$$K = \frac{v_F}{v_{out}} = \frac{R_2}{R_1 + R_2}$$

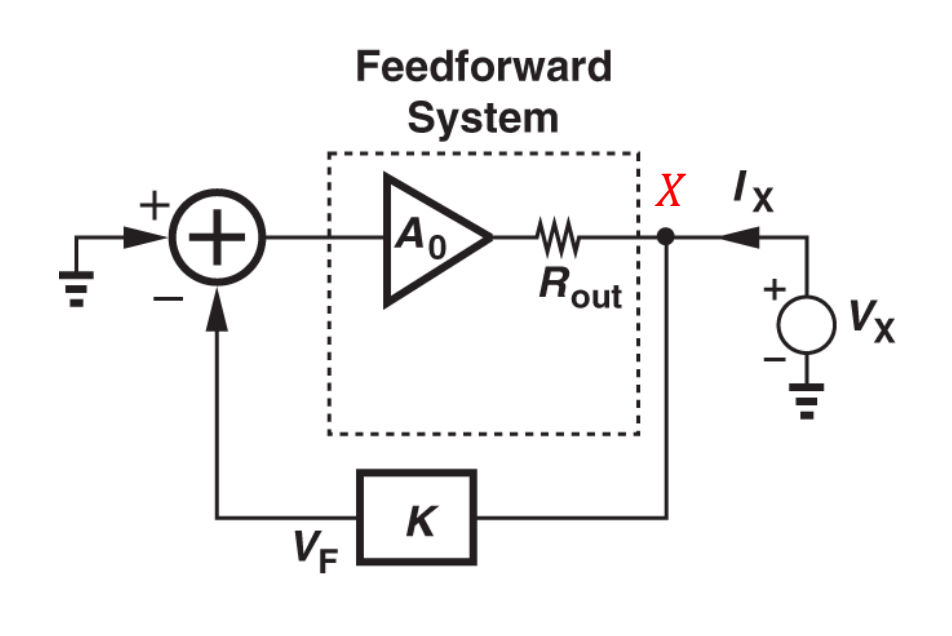
$$A_0 = g_m R_D$$

$$\frac{V_{in}}{I_{in}} = R_{in}(1 + KA_0)$$

$$= \frac{1}{g_m}\left(1 + \frac{R_2}{R_1 + R_2} g_m R_D\right)$$

Output Impedance of a V-V Feedback

- A better voltage source: the negative feedback lowers the output impedance



Apply KCL at node X,

$$I_X = \frac{V_X - A_0(-V_F)}{R_{out}}$$

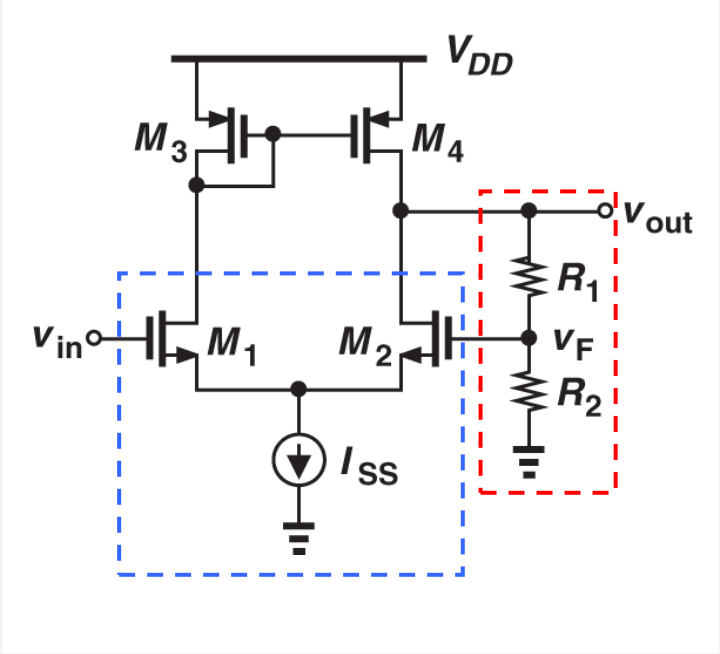
$$V_F = KV_X$$

$$I_X = \frac{V_X + A_0(KV_X)}{R_{out}} = \frac{(1 + KA_0)V_X}{R_{out}}$$

$$\frac{V_X}{I_X} = \frac{R_{out}}{(1 + KA_0)}$$

Example 12-17: Output Impedance of a V-V Feedback

- Calculate the output impedance of the stage shown in Fig. 12.36 if R_1+R_2 is very large.



Solution:

Recall from Example 12-15 that the open-loop output impedance is equal to

$$R_{out} = (r_{ON}||r_{OP})$$

and

$$K \cdot A_0 = \frac{R_2}{R_1 + R_2} \cdot g_{mN}(r_{ON}||r_{OP})$$

$$R_{out,closed} = \frac{(r_{ON}||r_{OP})}{1 + \frac{R_2}{R_1 + R_2} g_{mN}(r_{ON}||r_{OP})}$$

If the loop gain is much greater than unity,

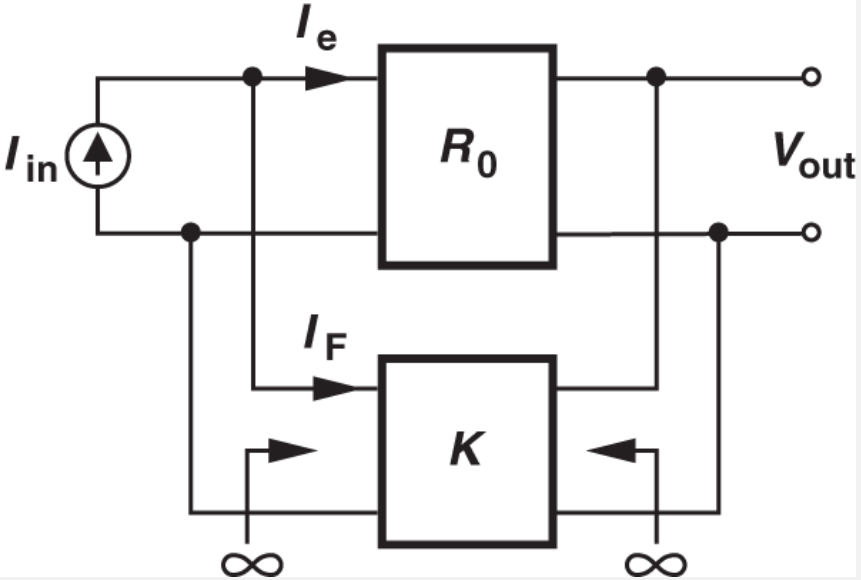
$$R_{out,closed} \approx \left(1 + \frac{R_1}{R_2}\right)\frac{1}{g_{mN}}$$

a value independent of r_{ON} and r_{OP}.

In other words, while the open-loop amplifier suffers from a high output impedance, the application of negative feedback lowers R_{out} to a multiple of $1/g_{mN}$.

Voltage-Current Feedback

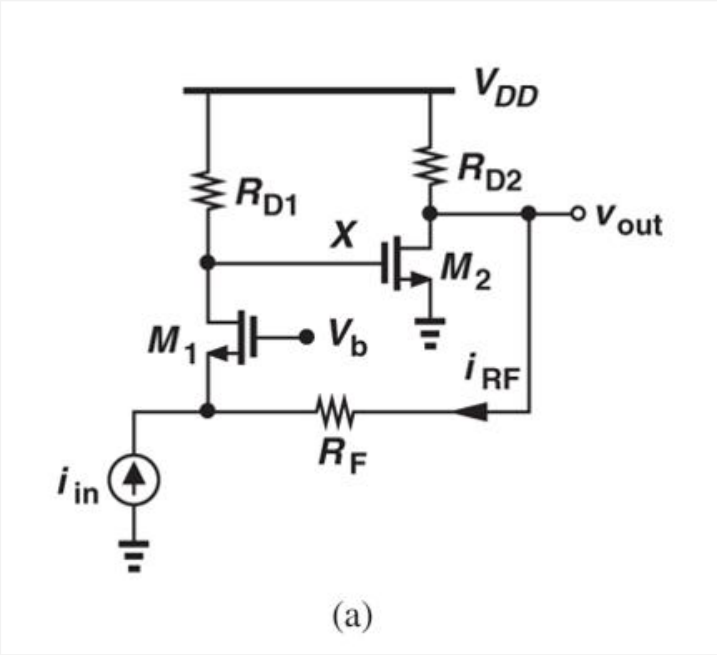
- In our terminology, the first term in "voltage-current feedback" refers to the quantity sensed at the output, and the second, to the quantity returned to the input. (This terminology is not standard.)
- This type is also called "shunt-shunt" feedback.



$$I_e = I_{in} - I_F$$
$$I_F = KV_{out}$$
$$V_{out} = I_e R_0$$
$$= (I_{in} - I_F)R_0$$
$$= (I_{in} - KV_{out})R_0$$
$$\frac{V_{out}}{I_{in}} = \frac{R_0}{1 + KR_0}$$

Example 12-18: Voltage-Current Feedback

- For the circuit shown in Fig. 12.38(a), assume $\lambda = 0$ and R_F is very large and (a) prove that the feedback is negative; (b) calculate the open-loop gain; (c) calculate the closed-loop gain.

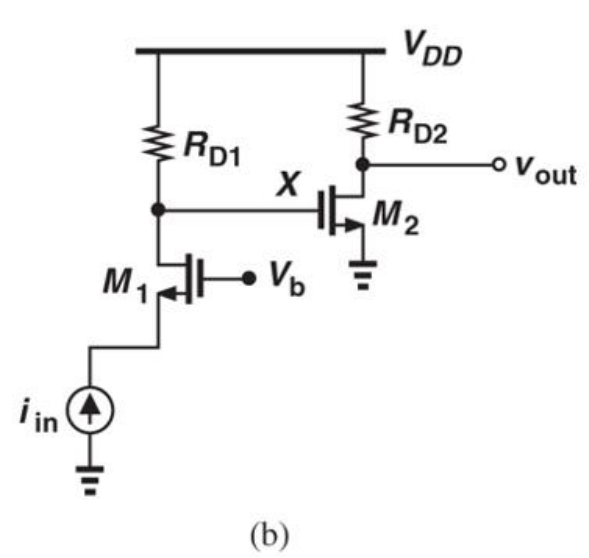


Solution: (a)

$$I_{in} \uparrow \Rightarrow I_{D1} \downarrow$$
$$\Rightarrow V_X \uparrow$$
$$\Rightarrow V_{out} \downarrow$$
$$\Rightarrow I_F \downarrow$$

Since the currents injected by I_{in} and I_{RF} into the input node change in opposite directions, the feedback is negative.

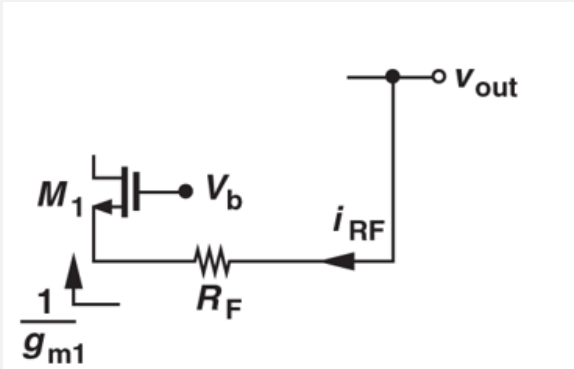
Solution: (b)



(b)

$$R_0 = \frac{V_{out}}{I_{in}}$$
$$= \frac{V_X}{I_{in}} \cdot \frac{V_{out}}{V_X}$$
$$= R_{D1} \cdot (-g_{m2} R_{D2})$$
$$= -g_{m2} R_{D1} R_{D2}$$

Solution: (c)

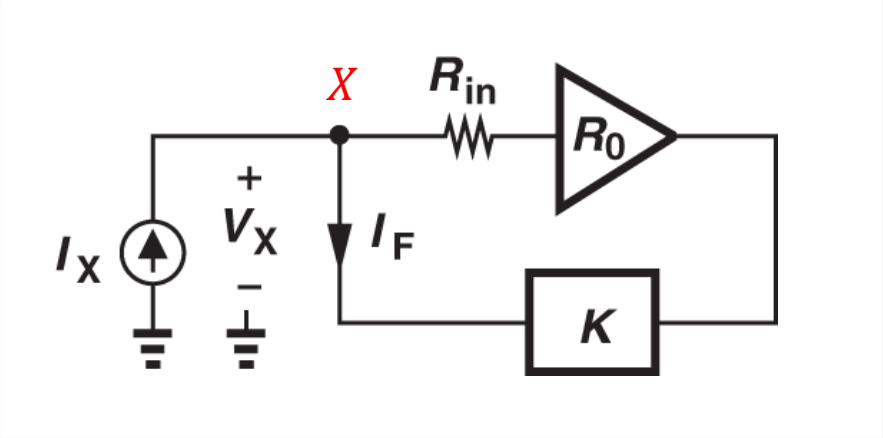


(c)

$$I_F = \frac{V_{out}}{R_F + \frac{1}{g_{m1}}}$$
$$= \frac{V_{out}}{R_F} \quad \left(\text{If } R_F \gg \frac{1}{g_{m1}}\right)$$
$$-I_F \triangleq K V_{out} \quad \Rightarrow \quad K = -\frac{1}{R_F}$$
$$R_{0,closed} = \left.\frac{V_{out}}{I_{in}}\right|_{closed} = \frac{-g_{m2} R_{D1} R_{D2}}{1 + \frac{g_{m2} R_{D1} R_{D2}}{R_F}}$$

Input Impedance of a V-C Feedback

- A better current sensor



$$I_F = K \cdot R_0 \cdot \frac{V_X}{R_{in}}$$

Apply KCL at node X,

$$I_X = \frac{V_X}{R_{in}} + I_F$$

$$= \frac{V_X}{R_{in}} + KR_0 \frac{V_X}{R_{in}}$$

$$= \frac{(1 + KR_0)V_X}{R_{in}}$$

$$R_{in,closed} = \frac{V_X}{I_X}$$

$$= \frac{R_{in}}{1 + KR_0}$$

Example 12-19: Input Impedance of a V-C Feedback

- Determine the closed-loop input impedance of the circuit studied in Example 12-18.

Solution:

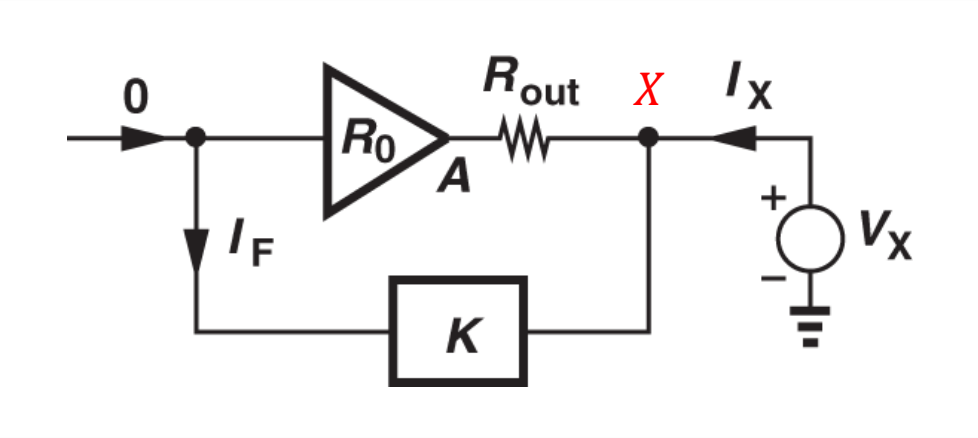
$$R_{in} = \frac{1}{g_{m1}} \quad \left(\text{assume } R_F \gg \frac{1}{g_{m1}}\right)$$

$$1 + KR_0 = \frac{1}{1 + \dfrac{g_{m2} R_{D1} R_{D2}}{R_F}} \quad (\text{Example } 12{-}18(\text{c}))$$

$$R_{in,closed} = \frac{1}{g_{m1}} \cdot \frac{1}{1 + \dfrac{g_{m2} R_{D1} R_{D2}}{R_F}}$$

Output Impedance of a V-C Feedback

- A better voltage source



$$I_F = KV_X$$
$$V_A = -I_F R_0$$

Apply KCL at node X,

$$I_X = \frac{V_X - V_A}{R_{out}}$$
$$= \frac{V_X + KV_X R_0}{R_{out}}$$
$$= \frac{(1 + KR_0)V_X}{R_{out}}$$

$$R_{out,closed} = \frac{V_X}{I_X}$$
$$= \frac{R_{out}}{(1 + KR_0)}$$

Example 12-20: Output Impedance of a V-C Feedback

- Calculate the closed-loop output impedance of the circuit studied in Example 12-18.

Solution:

$$R_{out} \approx R_{D2} \quad (\text{assume } R_F \gg R_{D2})$$

$$1 + KR_0 = \frac{1}{1 + \dfrac{g_{m2}R_{D1}R_{D2}}{R_F}} \quad (\text{Example } 12{-}18(\text{c}))$$

$$R_{out,closed} = \frac{R_{D2}}{1 + \dfrac{g_{m2}R_{D1}R_{D2}}{R_F}}$$

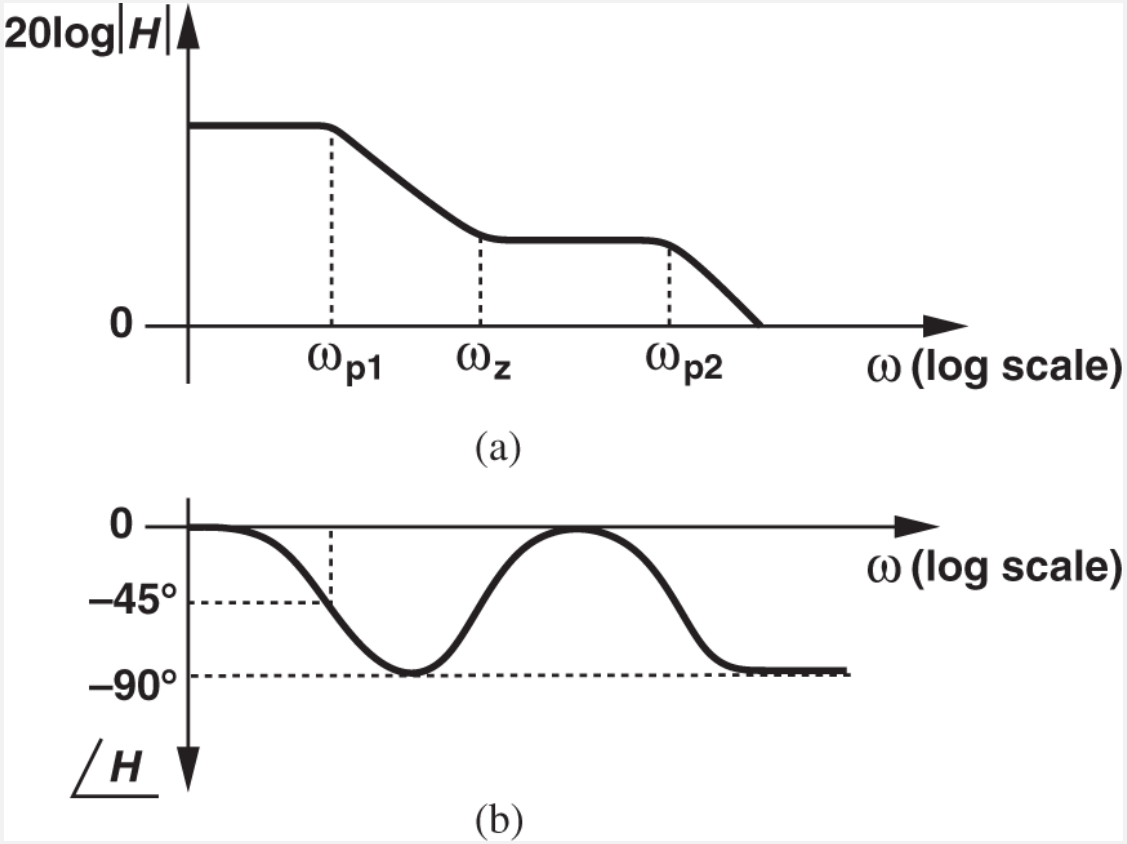
STABILITY IN FEEDBACK SYSTEMS

- Our studies in this chapter have thus far revealed many important benefits of negative feedback.
- Unfortunately, if designed poorly, negative-feedback amplifiers may behave “badly” or even oscillate. We say the system is marginally stable or simply unstable.
- In this section, we reexamine our understanding of feedback so as to define the meaning of “behaving badly” and determine the sources of instability.
- **Review of Bode's Rules**:

  The phase of a transfer function begins to decrease (increase) at one-tenth of the pole (zero) frequency, incurs a change of -45° (+45°) at the pole (zero) frequency, and experiences a total change of nearly -90° (+90°) at ten times the pole (zero) frequency.*

Example 12-34: Phase Response

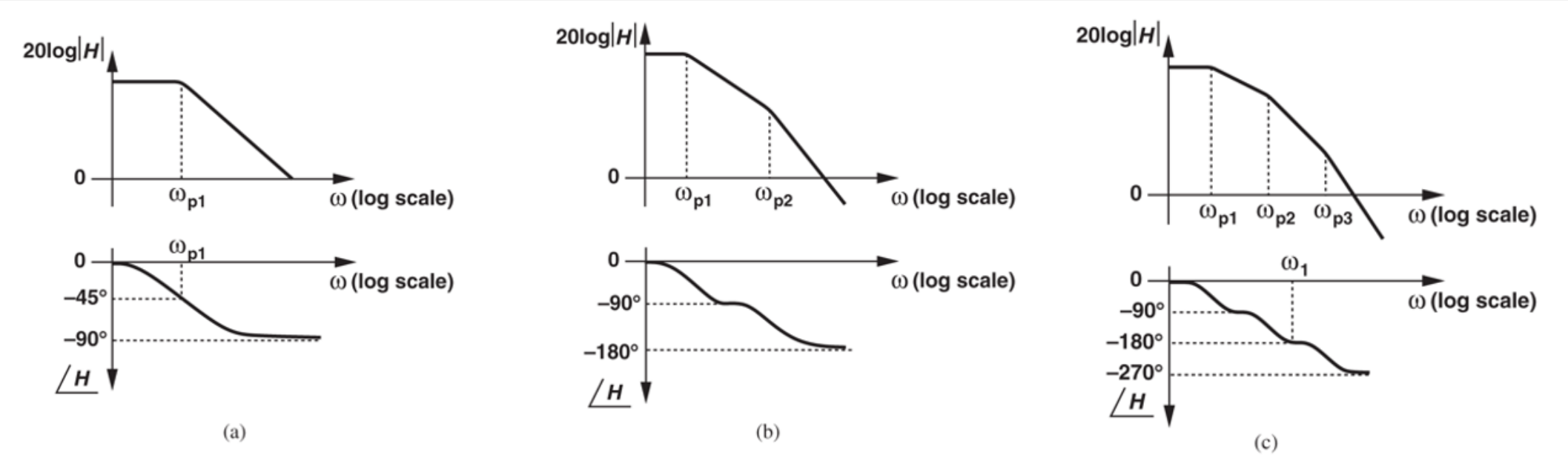
- Figure 12.63(a) depicts the magnitude response of an amplifier. Using Bode's rule, plot the phase response.1



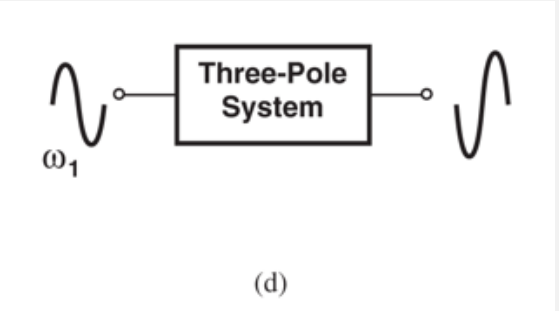
As it can be seen, the phase of H(jω) starts to drop at 1/10 of the pole, hits -45° at the pole, and approaches -90° at 10 times the pole.

Example 12-35: Three-Pole System

- Construct the magnitude and phase response of an amplifier having (a) one pole, (b) two poles, or (c) three poles.

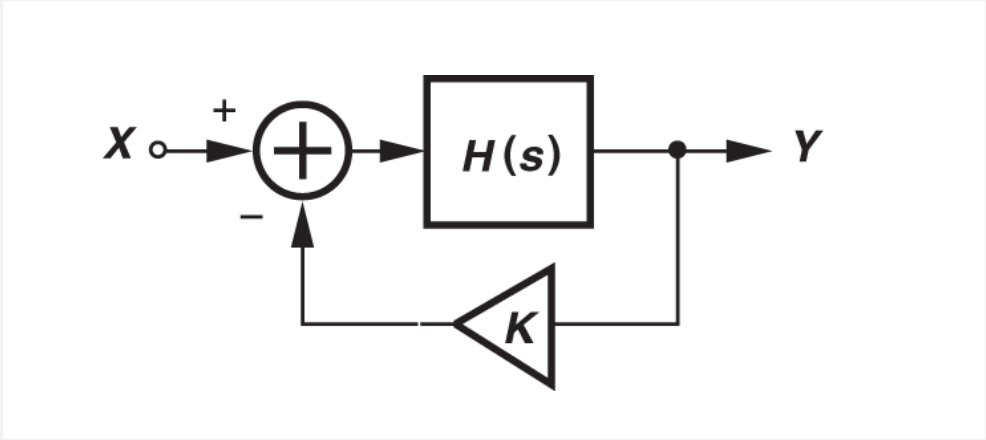


For a three-pole system, a finite frequency produces a phase of -180°, which means an input signal that operates at this frequency will have its output inverted.



Instability of a Negative Feedback Loop

- Substitute jω for s. If for a certain ω_1, $KH(j\omega_1)$ reaches -1, the closed loop gain becomes infinite. This implies for a very small input signal at ω_1, the output can be very large. Thus the system becomes unstable.



$$\frac{Y}{X}(s) = \frac{H(s)}{1 + KH(s)}$$

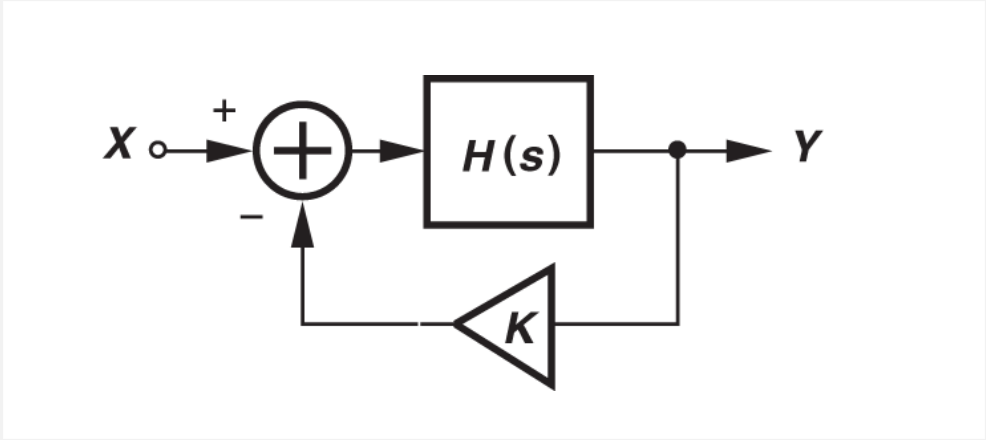
When $1 + KH(j\omega_1) = 0$, i.e.,

$$KH(j\omega_1) = -1$$

$$\frac{Y}{X}(j\omega_1) = \infty$$

“Barkhausen’s Criteria” for Oscillation

- For analysis purposes, we express the condition $KH(j\omega_1) = -1$ in a different form. Viewing KH as a complex quantity, we recognize that this condition is equivalent to



$$|KH(j\omega_1)| = 1$$
$$\angle KH(j\omega_1) = -180$$

Example 12-36

- Explain why a two-pole system cannot oscillate.

Solution:

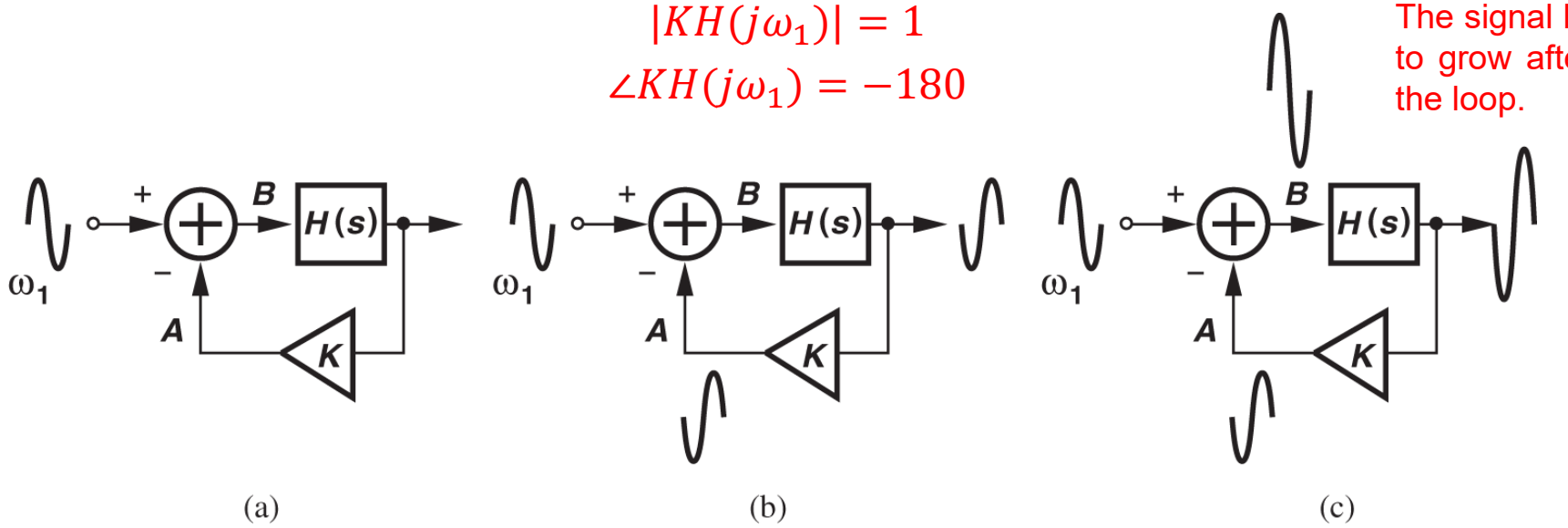
As evident from the Bode plots in Fig. 12.64(b), the phase shift produced by such a system reaches -180° only at $\omega = \infty$, where $|H| \rightarrow 0$.

In other words, at no frequency are both Eqs. (12.185) and (12.186) satisfied.

Time Evolution of Instability

$$|KH(j\omega_1)| = 1$$
$$\angle KH(j\omega_1) = -180$$

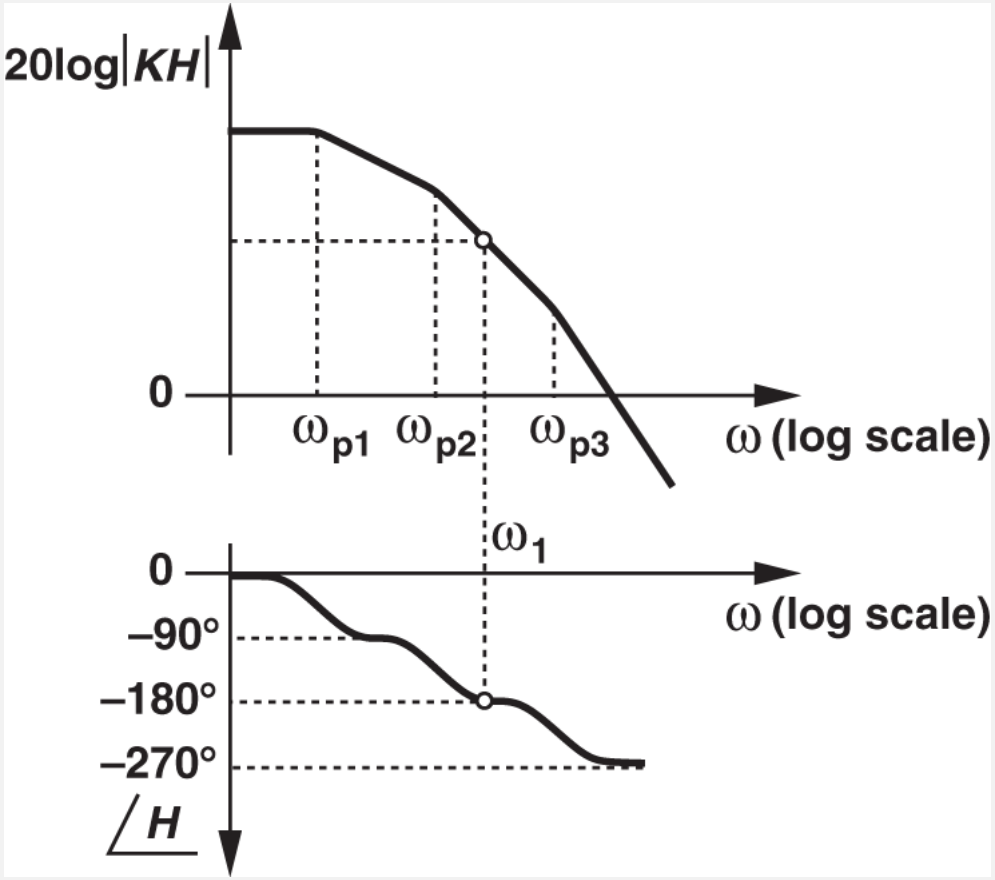
The signal level thus continues to grow after each trip around the loop.



Evolution of oscillatory system with time: (a) a component at is sensed at input, (b) the component returns to subtractor with a 180° phase shift, and (c) the subtractor enhances the signal at node B.

Example 12-37: Oscillation

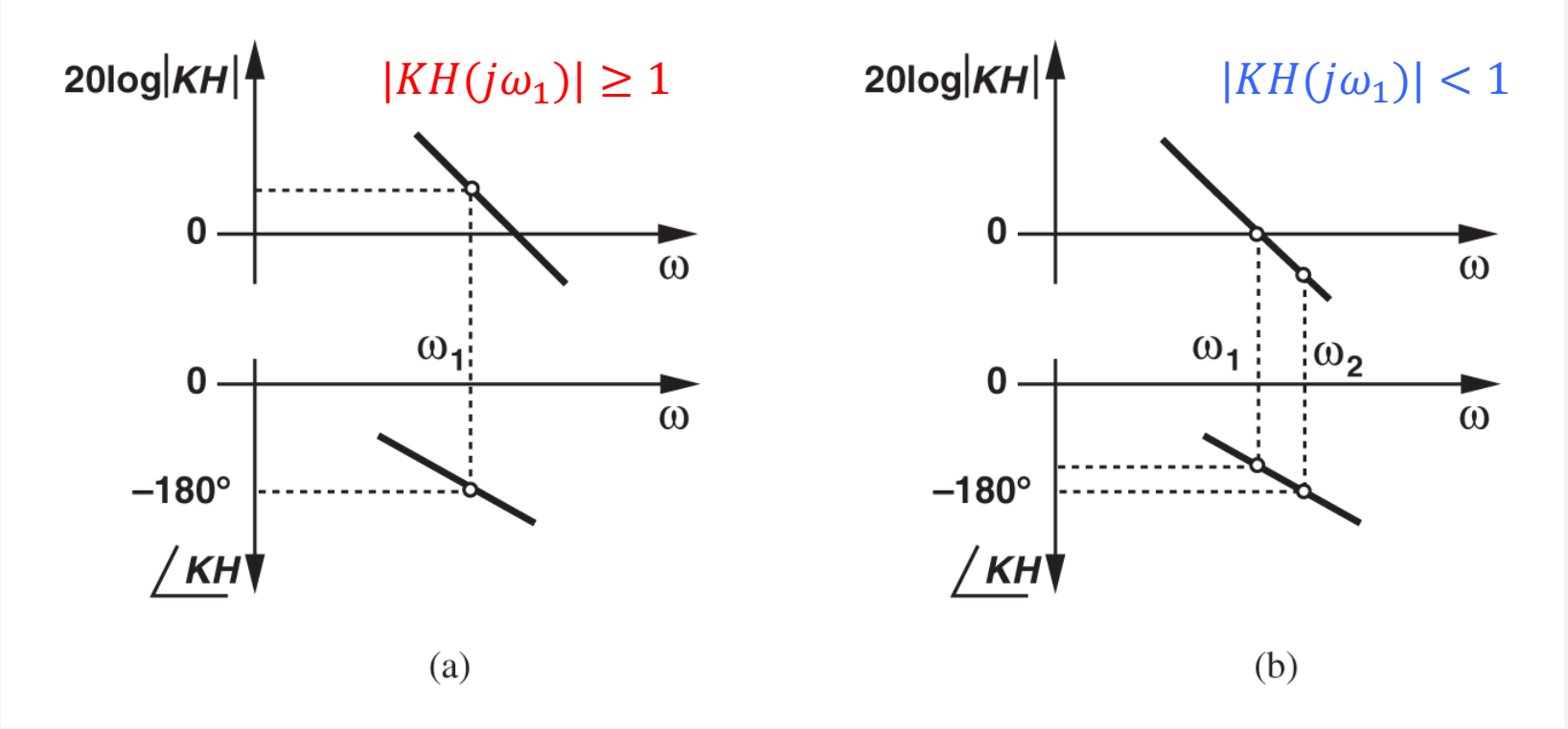
- A three-pole feedback system exhibits the frequency response depicted in Fig. 12.67. Does this system oscillate?



This system oscillates, since there's a finite frequency at which the phase is -180° and the gain is greater than unity. In fact, this system exceeds the minimum oscillation requirement.

Condition for Oscillation

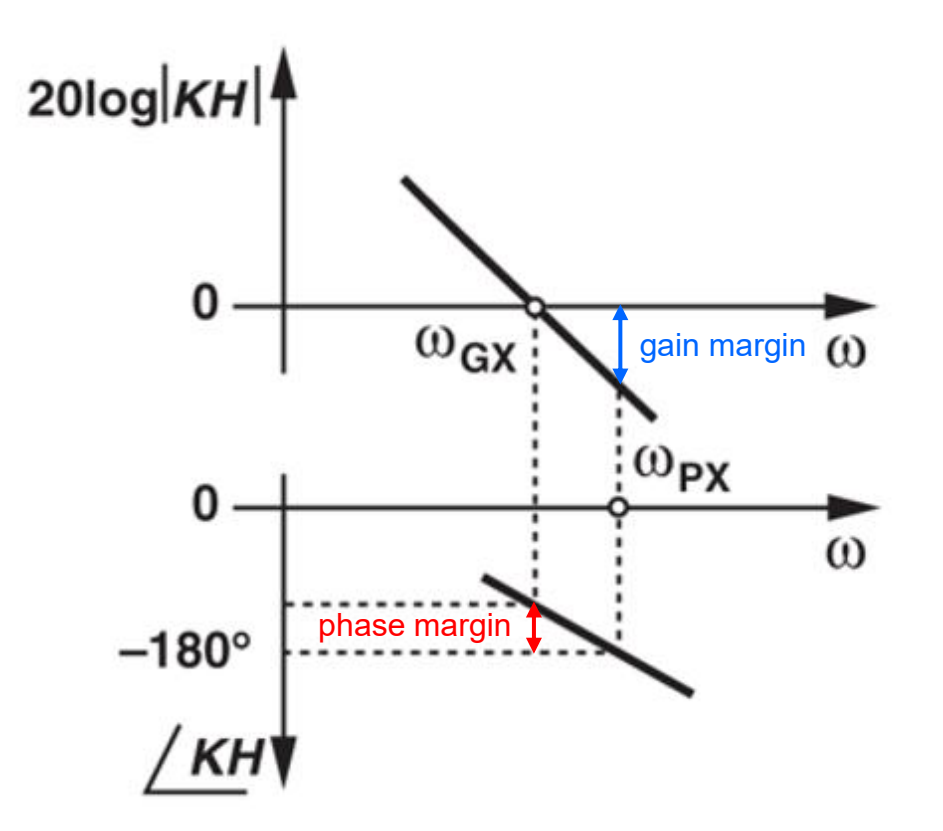
- Although for both systems above, the frequencies at which |KH| = 1 and ∠KH = -180° are different, the system on the left is still unstable because at ∠KH = -180°, |KH| > 1. Whereas the system on the right is stable because at ∠KH=-180°, |KH|<1.



Systems with phase reaching -180° (a) before and (b) after the loop gain reaches unity.

Phase Margin

- A measure commonly used to quantify the stability of feedback systems is the "phase margin" (PM). The more stable a system is, the greater is the difference between $\angle H(\omega_{GX})$ and -180°. Indeed, this difference is called the phase margin:



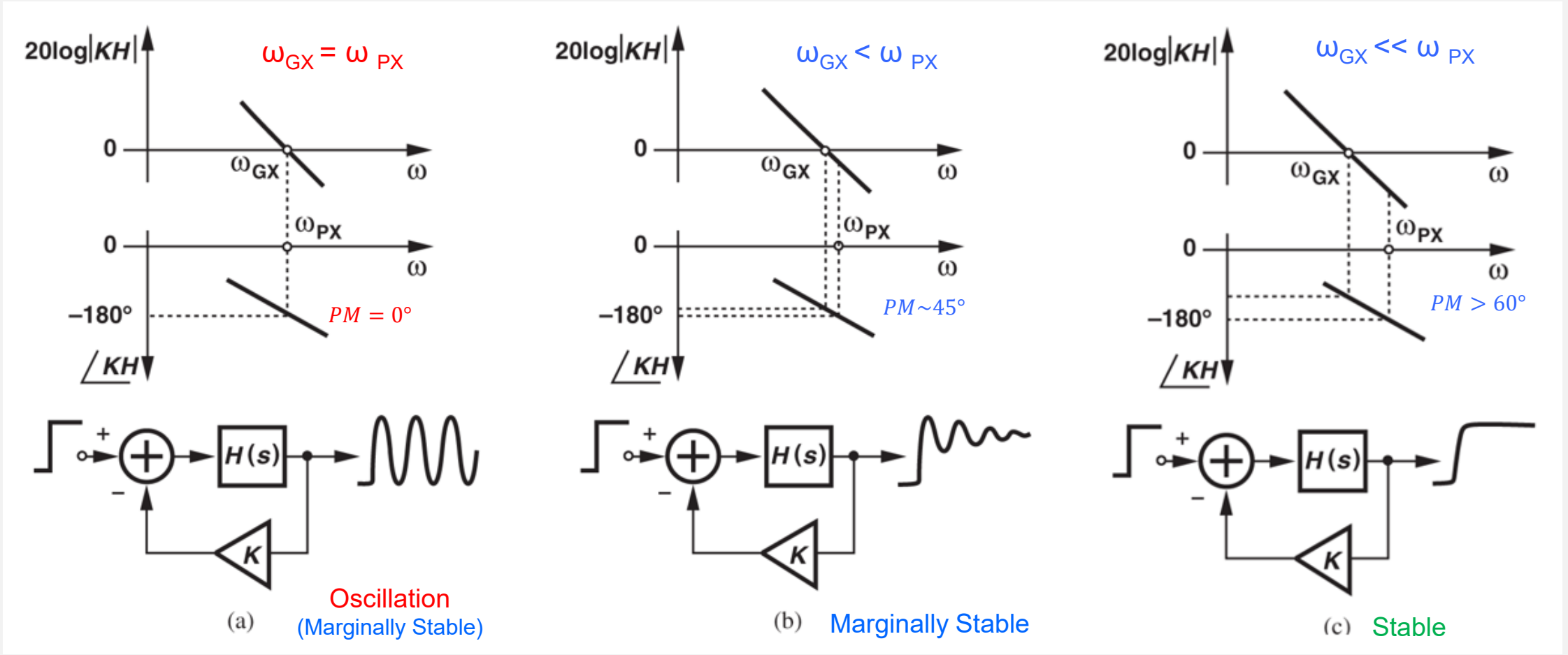
ω_{PX}: ("phase crossover") is the frequency at which $\angle KH = -180^{o}$.

ω_{GX}: ("gain crossover") is the frequency at which $|KH| = 1$.

$$\begin{aligned} \text{PM} &= \angle H(\omega_{GX}) - (-180) \\ &= \angle H(\omega_{GX}) + 180 \end{aligned}$$

Condition for Stability

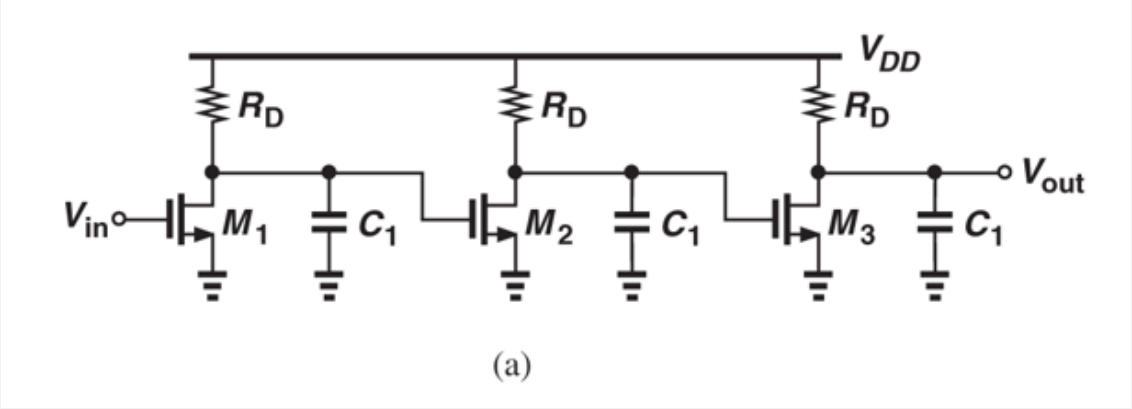
- **Condition for Stability**: $\omega_{GX} < \omega_{PX}$



Systems with (a) coincident gain and phase crossovers, (b) gain crossover slightly below phase crossover, and (c) gain crossover well below phase crossover.

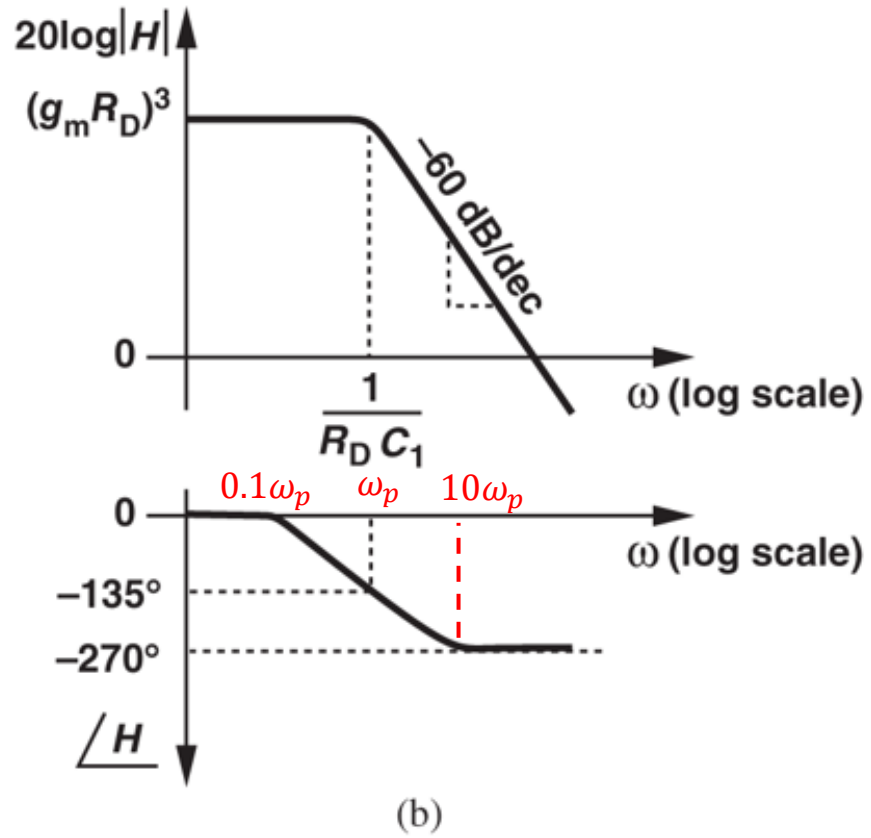
Example 12-38

- We wish to apply negative feedback with K = 1 around the three-stage amplifier shown in Fig. 12.69(a). Neglecting other capacitances and assuming identical stages, (a) plot the frequency response of the circuit and (b) determine the condition for stability. Assume $\lambda = 0$.



(a)

Solution: (a)

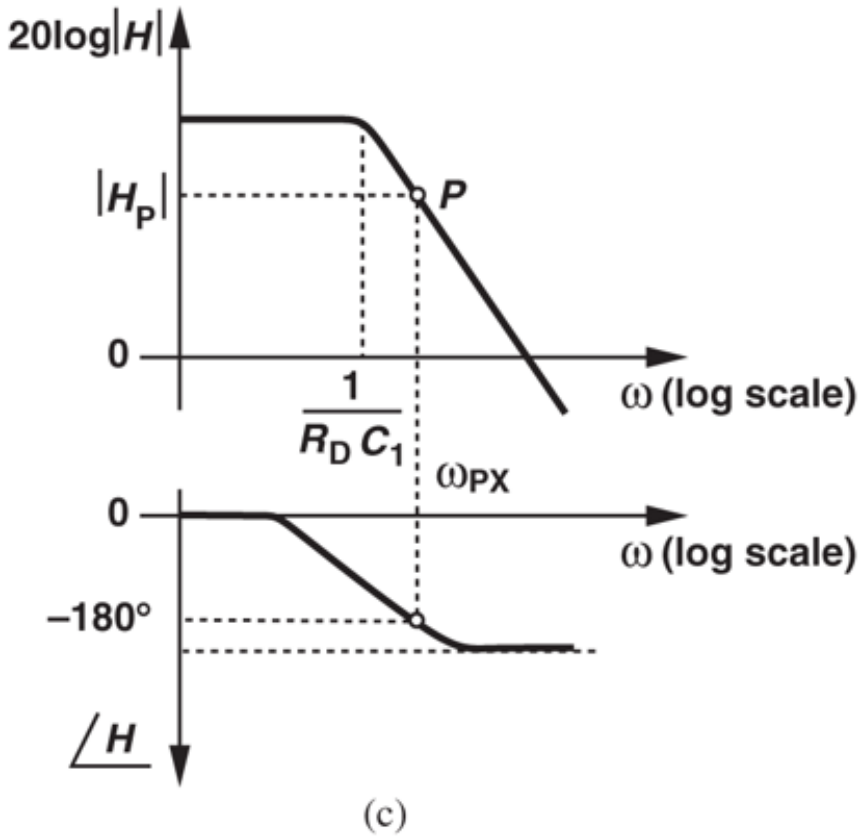


(b)

The system has a third-order pole at

$$\omega_p = \frac{1}{R_D C_1}$$

Solution: (b)



(c)

$\because K = 1 \Rightarrow |KH| = |H|$

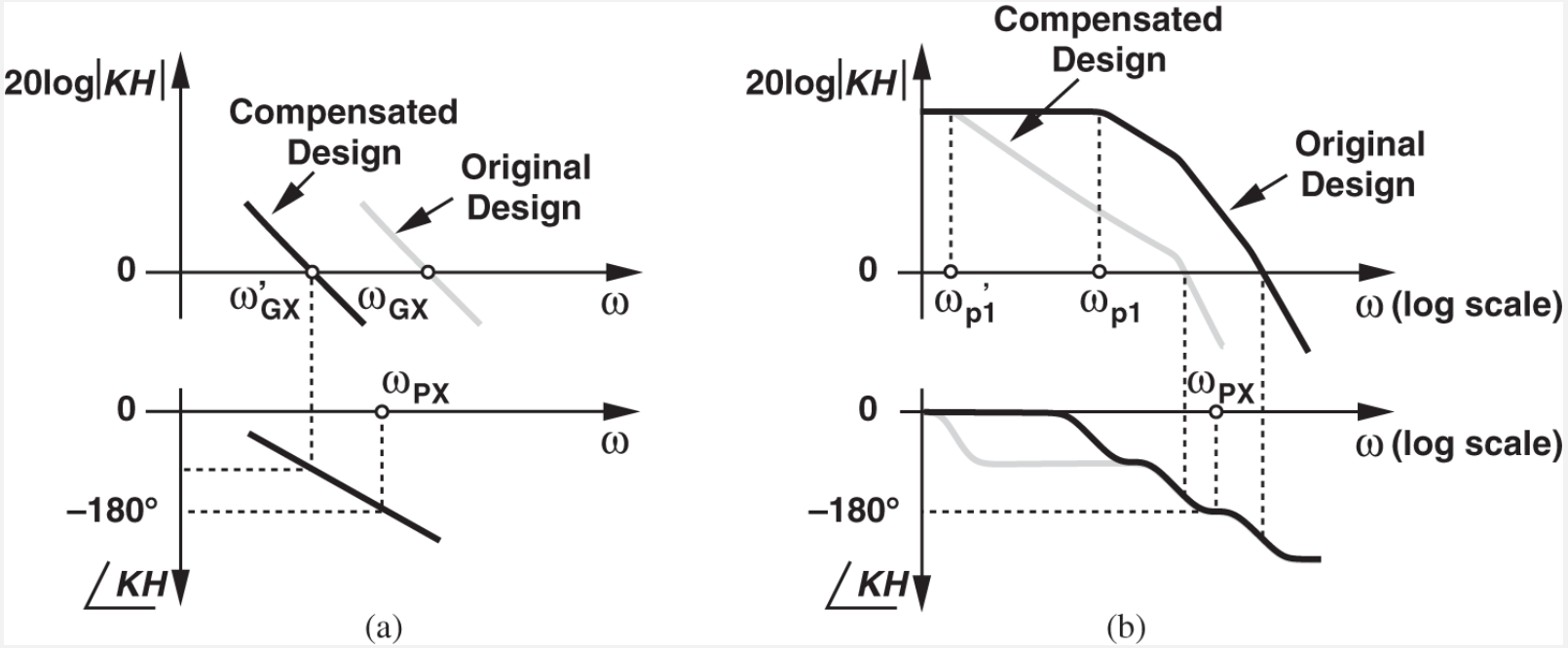
$|H(\omega_{PX})| < 1$

guarantees that the unity-feedback system is stable, i.e.,

$|H_p| < 1, K = 1$.

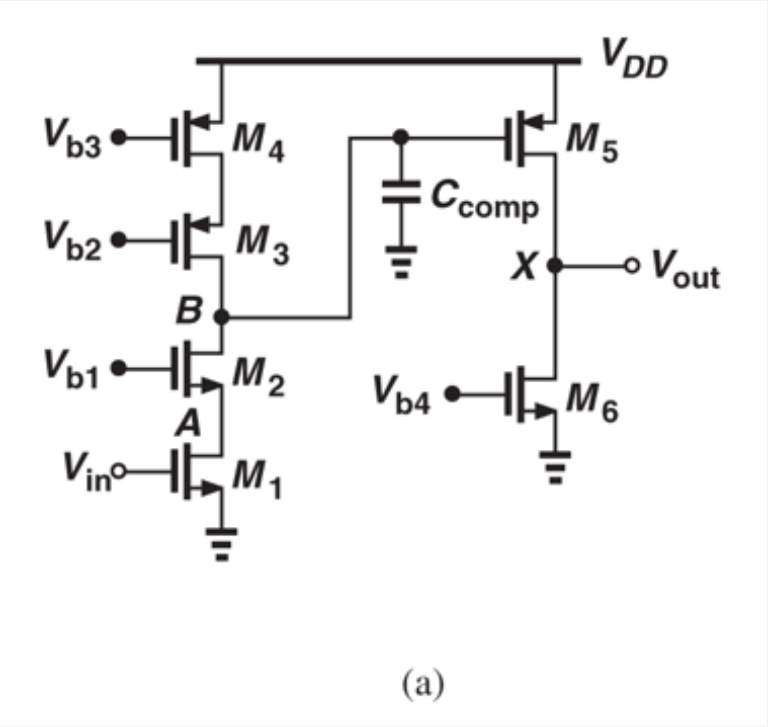
Frequency Compensation

- It is possible that after the design of an amplifier is completed, the phase margin proves inadequate. How is the circuit modified to improve the stability? For example, how do we make the three-stage amplifier of Example 12-38 stable if K = 1 and gm_{RD} > 2? The solution is to make two of the poles unequal in magnitude.
- **Frequency compensation**: By moving ω_{GX} closer to the origin while maintaining ω_{PX} unchanged, the phase margin can be improved.

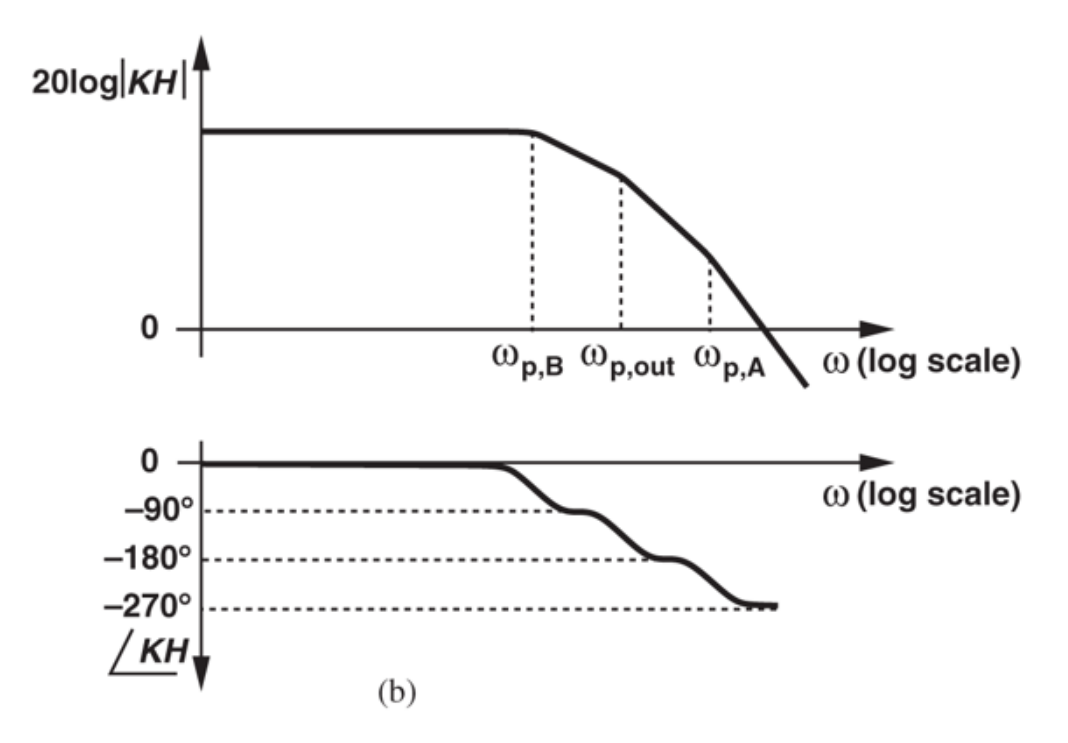


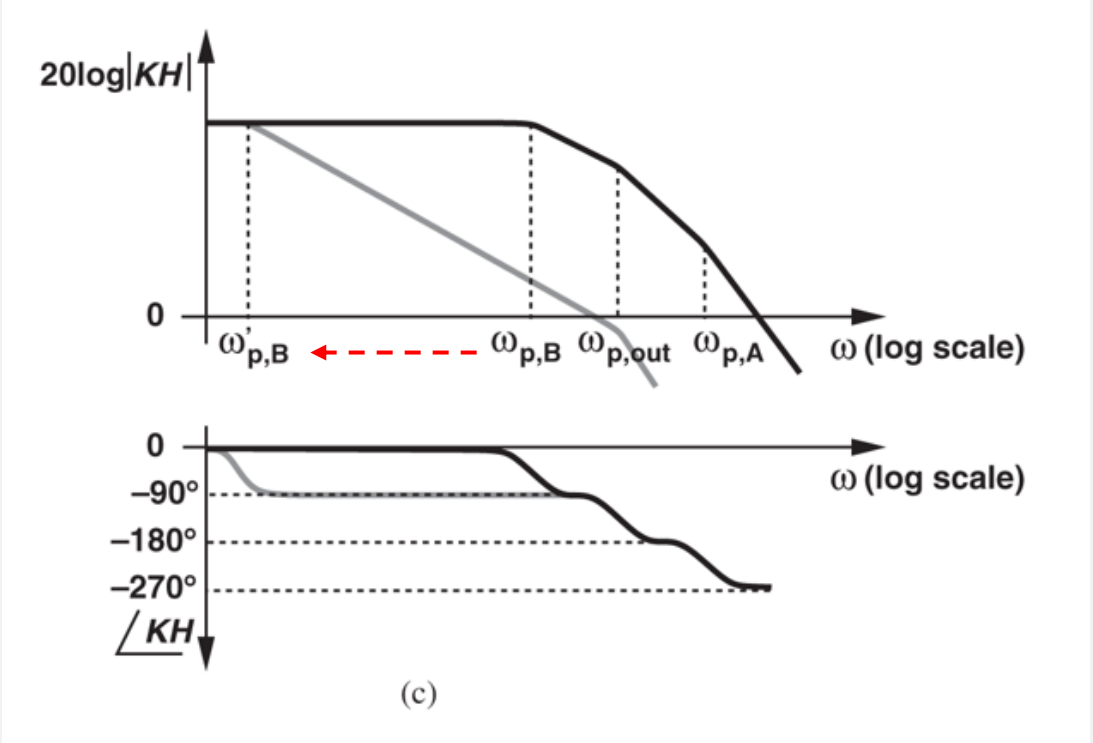
Example 12-42: Frequency Compensation

- The amplifier shown in Fig. 12.75(a) employs a cascode stage and a CS stage. Assuming that the pole at node B is dominant, sketch the frequency response and explain how the circuit can be "compensated."



Solution:

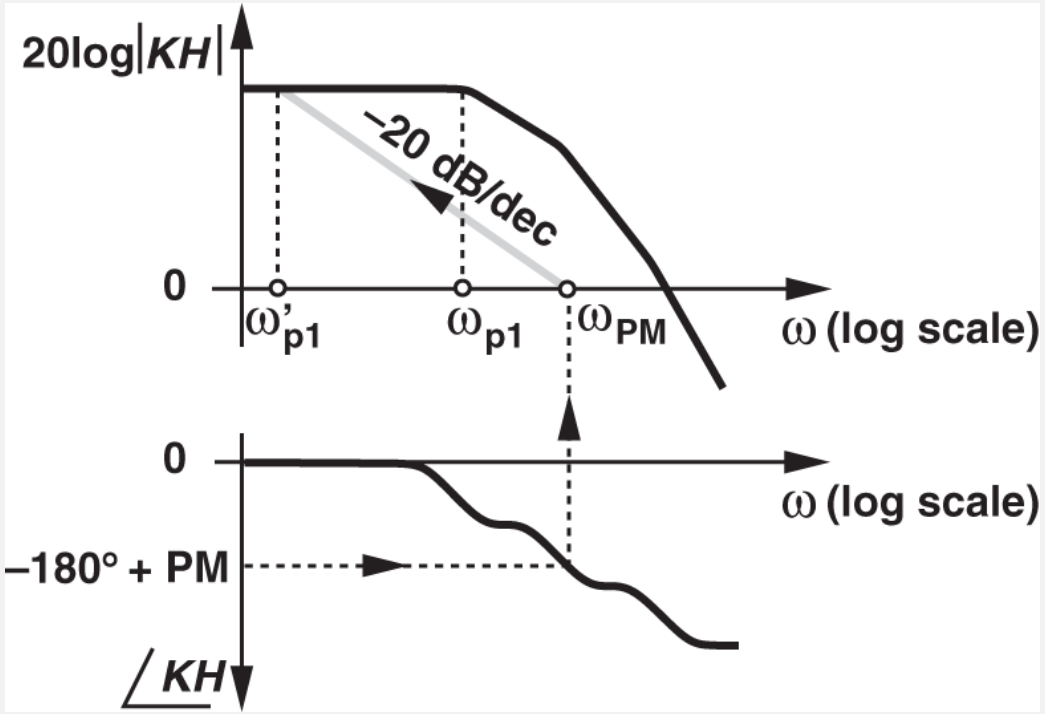




C_{comp} is added to lower the dominant pole so that ω_{GX} occurs at a lower frequency than before, which means phase margin increases.

Systematic Method for Frequency Compensation

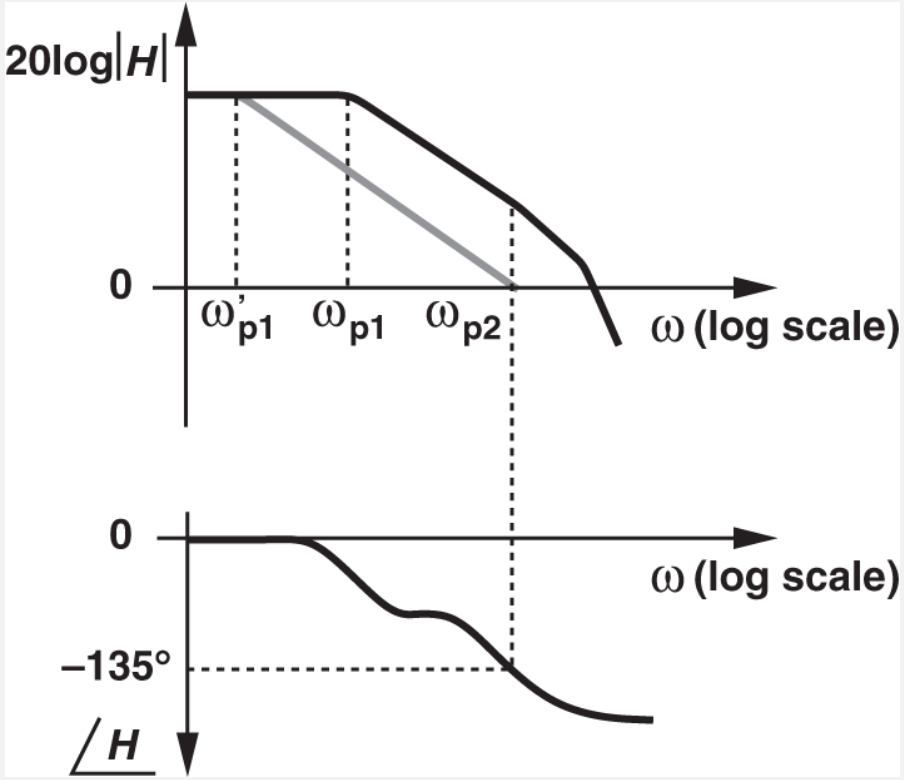
- **Frequency Compensation Procedure**:
  1) We identify a PM, then -180°+PM gives us the new ω_{GX}, or ω_{PM}.
  2) On the magnitude plot at ω_{PM}, we extrapolate up with a slope of +20dB/dec until we hit the low frequency gain then we look "down" and the frequency we see is our new dominant pole, ω_P'.



Example 12-43: 45° Phase Margin Compensation

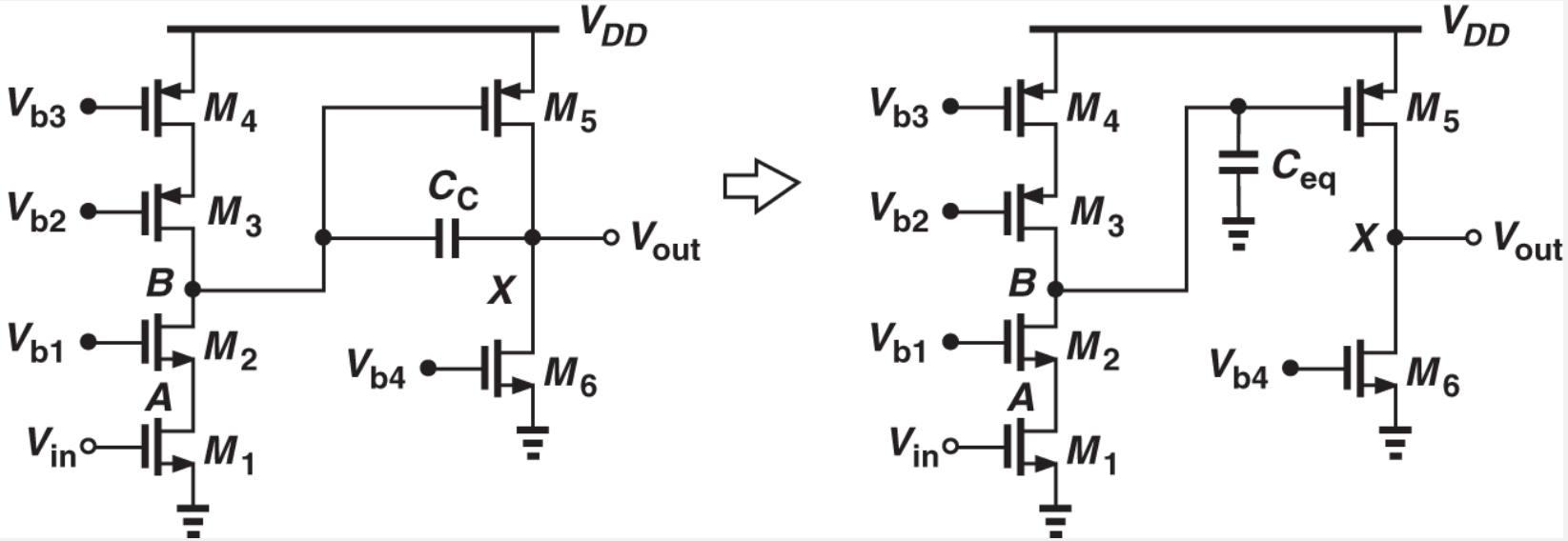
- A multipole amplifier exhibits the frequency response plotted in Fig.12.77(a). Assuming the poles are far apart, compensate the amplifier for a phase margin of 45° with K = 1.

Solution:



Miller Compensation

- In Example 12-42, we noted that a capacitor can be tied from node B to ground to compensate the amplifier. The required value of this capacitor may be quite large, necessitating a large area on an integrated circuit.
- To save chip area, Miller multiplication of a smaller capacitance creates an equivalent effect.



$$C_{eq} = [1 + g_{m5}(r_{O5}||r_{O6})]C_c$$